



Intrapartum Fetal Evaluation

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KEY ABBREVIATIONS

Amnioinfusion	AI	Fetal electrocardiography	FEC
Electronic fetal monitoring	EFM	Fetal scalp electrode	FSE
Eunice Kennedy Shriver National Institute of Child Health and Human Development	NICHD	Intrauterine pressure catheter	IUPC
		Tocodynamometer	TOCO

BACKGROUND

Intrapartum fetal monitoring to assess fetal well-being during labor and delivery is a key component of intrapartum management. In fact, more than 80% of laboring patients in the United States undergo intrapartum electronic fetal monitoring (EFM).¹ “Non-reassuring” fetal status as interpreted on the basis of EFM accounts for nearly one-quarter of primary cesarean deliveries.² This ubiquitous form of intrapartum fetal assessment has evolved from its origins as intermittent auscultation of fetal heart tones to what is now most commonly continuous EFM. Fetal heart rate (FHR) characteristics

reflect the fetal response to intrauterine hypoxia.³ The theorized benefit of intrapartum EFM is to identify and intervene on fetal hypoxia secondary to the physiologic demands of normal and abnormal labor. Normal components of labor, such as contractions throughout the first and second stages and maternal expulsive efforts during the second stage, result in transient decreases in oxygen delivery and gas exchange for the fetus, which places the fetus at risk for interruption in oxygenation and hypoxia even in uncomplicated pregnancies (Fig. 17.1).^{4–6} Prolonged, compromised oxygen delivery to the fetus may result in neonatal acidemia and associated morbidity. Although evidence for the prevention of such morbidity

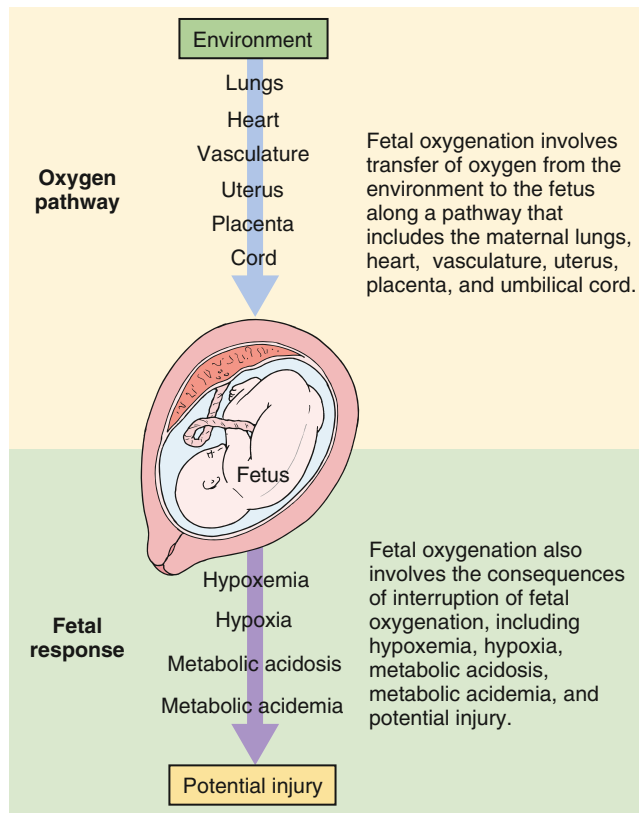


Fig. 17.1 The Oxygen Pathway.

with continuous EFM is modest, intrapartum fetal monitoring remains an ingrained component of modern-day obstetrics.

FETAL HEART RATE MONITORING

Normal Physiologic Control of the Fetal Heart Rate

The FHR and pattern are controlled by an interplay of the sympathetic and parasympathetic nervous systems, with central and peripheral influences. The first medical documentation of fetal heart tones was made in 1818 by the Swiss physician Dr. Francois-Isaac Mayor, who performed fetal auscultation by placing his ear next to the maternal abdomen. The introduction of the fetal stethoscope in the early 1900s allowed physicians to slowly link patterns recognized during intermittent auscultation with adverse neonatal outcomes, ultimately establishing criteria for fetal distress.⁷ The standard of care before the introduction of EFM was a combination of intermittent auscultation and fetal scalp pH sampling.

Intermittent Auscultation

Intermittent auscultation was accomplished by placing a noninvasive device on the maternal abdomen and interpreting the FHR pattern by sound. A variety of devices were used over the years, including auscultative devices similar to the stethoscope, as well as Doppler devices that detected movement of the fetal heart. These approaches offered the opportunity to intermittently interpret the FHR pattern but did not allow for continuous assessment. Given the ongoing, periodic, and often unpredictable nature of the physiologic challenges of labor, the inability to continuously monitor the fetus and the possibility that fetal distress could go undetected was the impetus for continuous fetal monitoring.

Electronic Fetal Monitoring

In 1958, Hon and Kubli published the first description of the use of EFM in humans. They reported that patterns observed in 80 laboring women were associated with fetal acidemia as measured with scalp sampling or umbilical cord assessment at delivery.⁸ The introduction of EFM in the 1960s⁹ led to the contemporary and commercially available monitoring systems that either use noninvasive Doppler-based technology for transabdominal FHR recognition or invasive fetal electrocardiography (ECG) via an electrode placed on the fetal scalp.^{10,11} Following the early descriptions and widespread availability of this modality, 13 randomized trials were conducted to compare the use of intrapartum EFM with intermittent auscultation aimed at reducing neonatal morbidity and improving obstetric outcomes (Table 17.1).^{12–24} Among these efforts, a large trial of 12,964 women, each of whom was cared for by a single provider throughout labor either with intermittent auscultation or EFM, showed a reduction in neonatal seizures among the infants whose mothers were managed with EFM.²⁵ However, after 4 years, there was no difference between the groups with regard to infant outcomes.²⁶ There were no other neonatal differences between the groups, including no differences in the rates of cerebral palsy. The remainder of the trials provided no evidence for benefit of EFM over intermittent auscultation to improve neonatal outcomes. A systematic review of these trials that included more than 37,000 women found no evidence for improved neonatal outcomes with EFM compared with intermittent auscultation, but it showed a significant increase in cesarean delivery among patients managed with EFM.¹²

After the use of intrapartum EFM in the United States became widespread, Nelson and colleagues²⁷ performed a nested case-control study among 150,000 live singleton births weighing greater than 2500 g to examine the effectiveness of EFM for the reduction of cerebral palsy. The patterns of 78 infants with cerebral palsy were compared with those of 300 randomly selected controls. The authors found that multiple late decelerations (odds ratio [OR], 3.9; 95% confidence interval [CI], 1.7 to 9.3) and decreased variability (OR, 2.7; 95% CI, 1.1 to 5.8) were associated with moderate to severe cerebral palsy but that most fetuses with those patterns did not develop cerebral palsy, with a false-positive rate of 99.8%. The authors also found that cesarean sections were often performed in the setting of these patterns and that, if this was common practice, many such surgeries would be performed without benefit to the fetus but with increased maternal risks.

With the widespread use of intrapartum EFM, subsequent population-based observational studies have reported more optimistic outcomes. In 2004, Chen and colleagues¹ examined the association between the use of intrapartum EFM and neonatal and infant deaths using the linked US birth and death data. They reported that 89% of the women studied underwent intrapartum EFM and that its use was associated with a significant reduction in infant mortality (adjusted relative risk [aRR], 0.75; 95% CI, 0.69 to 0.81), which was driven predominantly by a reduction in early neonatal death (aRR, 0.50; 95% CI, 0.44 to 0.57).¹ Another population-based cohort study of more than 55 million births of infants without anomalies between 24 and 44 weeks' gestation examined the impact of intrapartum EFM on US trends in adverse perinatal outcomes between 1990 and 2004.²⁸ These authors found that a 17% increase in the use of intrapartum EFM was associated with a 5% decrease in early neonatal deaths and a 2% decrease in late neonatal deaths, as well as an increase over the same period of cesareans and operative vaginal deliveries performed for concerns of fetal well-being. **Taken together, although the original trials and subsequent meta-analyses did not offer evidence that**

TABLE 17.1 Randomized Trials of Electronic Fetal Monitoring (EFM) Versus Intermittent Auscultation

Study	Year	Number	Perinatal Morbidity	Perinatal Mortality	Cesarean Rate
Haverkamp ¹³	1976	483	No difference	No difference	Higher in EFM group
Renou ¹⁴	1976	350	See footnote ^a	No difference	Higher in EFM group
Kelso ¹⁵	1978	504	No difference	No difference	Higher in EFM group
Haverkamp ¹⁶	1979	690	No difference	No difference	Higher in EFM group
Wood ¹⁷	1981	989	No difference	No difference	No difference
MacDonald ¹⁸	1985	12,964	Lower in EFM group ^b	No difference	No difference
Neldam ¹⁹	1986	969	No difference	No difference	No difference
Leveno ²⁰	1986	14,618	No difference	No difference	Not reported ^c
Luthy ²¹	1987	246	No difference	No difference	No difference
Vintzileos ²²	1993	1428	No difference	Lower in EFM group ^d	No difference
Herbst ²³	1994	4044	No difference	No difference	No difference
Madaan ²⁴	2006	100	No difference	No difference	No difference

^aLower rates of neurologic signs and symptoms in the EFM group but differences in frequency of neonatal seizures did not reach statistical significance.

^bFewer neonatal seizures were reported in the EFM group, but no difference was found at 4 years of age.

^cThe cesarean rate for fetal indications was higher in the EFM group, but overall cesarean rates were not reported.

^dThe study ended after the third quarterly review because of a statistically significant fivefold decrease in perinatal mortality in the EFM group.

intrapartum EFM was superior to intermittent auscultation for preventing neonatal morbidity—perhaps due in part to the quality of some of the studies—the ubiquitous nature of intrapartum EFM use in the past two decades has enabled epidemiologic studies suggesting possible neonatal benefit. Nevertheless, studies have also demonstrated that the number of operative deliveries that would have to be performed to prevent one adverse perinatal outcome would be high, which is consistent with the pervasive findings that the use of intrapartum EFM is associated with an increase in cesarean and operative vaginal deliveries.^{12,26,29,30}

ELECTRONIC FETAL MONITORING TECHNOLOGY

External Monitors

EFM is accomplished noninvasively with a device placed on the maternal abdomen kept in place by a flexible strap (Fig. 17.2). The device contains a fetal Doppler transducer that detects fetal cardiac motion or the motion of a major vessel. Because the monitor captures movement and is on the maternal surface, several factors—such as fetal or maternal movement or maternal body wall thickness—can interrupt the signal.

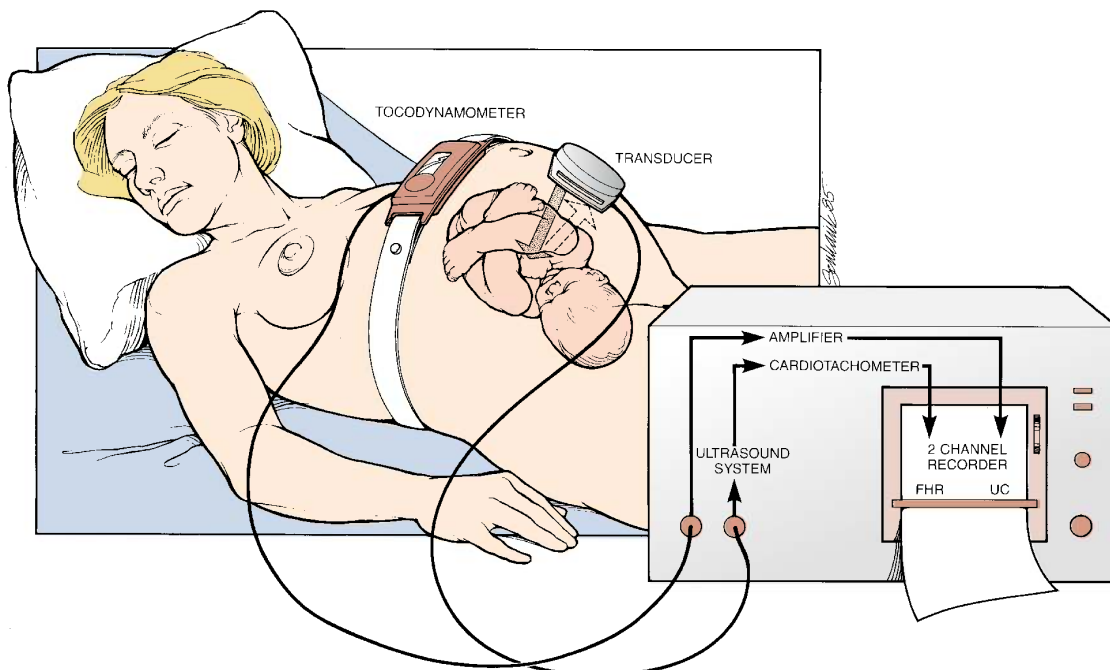


Fig. 17.2 Instrumentation for External Monitoring. Fetal heart rate (FHR) is monitored using the Doppler ultrasound transducer, which both emits and receives the reflected ultrasound signal that is then counted and recorded. Uterine contractions (UC) are detected by the pressure-sensitive tocodynamometer and are then amplified and recorded.

Simultaneously, a second similar Doppler device, called a tocodynamometer, monitors uterine activity. Importantly, the tocodynamometer can demonstrate the presence and duration of contractions. The tocodynamometer, however, cannot measure the strength of the contractions. Both the FHR monitor and the tocodynamometer are connected to a bedside terminal that plots the heart rate patterns in real time on EFM paper (moving at 3 cm/min in the United States and 1 cm/min in Europe and other parts of the world) and/or plotted virtually on an electronic display, allowing providers to visualize and interpret the patterns at bedside.

Internal Monitors

Sometimes acquisition of the FHR and/or contraction patterns by internal monitors becomes important for clinical interpretation. This can occur frequently when the maternal body habitus limits the ability of external monitors to capture a signal reliably because the monitors are based on ultrasound technology. **The FHR can be internally monitored with a fetal scalp electrode (FSE), a small metal corked electrode that is placed on the skin of the fetal scalp.** The other end of the cord connects to the same monitor used for external monitoring, generating the same output (Fig. 17.3). The input, as opposed to fetal heart motion that is captured and transmitted by external monitors, is the electrical activity of the fetal heart measured by the interval between beats (R-R interval). The advantage of internal fetal monitors is that they remove the possibility of interference with the signal from maternal motion, body habitus, and so on. Thus the signal is more accurate and continuous. Because use of the scalp electrode is invasive, it is not appropriate for all clinical scenarios and is not without some risk. In the setting of maternal infections, such as human immunodeficiency virus or hepatitis, FSEs are avoided to reduce the risk of vertical transmission. In addition, though relatively rare,

there have been reports of neonatal complications such as hematomas and scalp abscesses.^{31,32}

An invasive approach to uterine contraction monitoring is also clinically available. An intrauterine pressure catheter (IUPC), which is a flexible catheter with a pressure transducer at the tip, can be advanced through the cervix into the uterus, provided that the membranes are ruptured. Instead of the ultrasound-based technology used for the external tocodynamometer, the catheter measures the uterine pressure directly. The advantage of this approach is that it allows the clinician to gauge both the frequency of contractions and their strength. As with the difference between external and internal monitors for gauging the FHR, use of an IUPC overcomes the maternal factors that could interrupt the external signal. However, risks associated with the use of IUPCs have also been demonstrated. Harper and colleagues,³³ in a retrospective cohort study comparing the use of IUPCs with external monitoring, found that women monitored with IUPCs had a higher risk of maternal fever and infection, which was similar to the findings reported by Soper and colleagues.³⁴ **FSEs and IUPCs are commonly used, relatively safe clinical tools to assist in the assessment of FHR patterns and contractions; however, given the rare but possible associated complications, they should not be used without clinical indications.**

NOMENCLATURE

Current Terminology

In 2008, a consensus conference was convened by the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD); the American College of Obstetrics and Gynecology (ACOG); the Society for Maternal-Fetal Medicine (SMFM); the

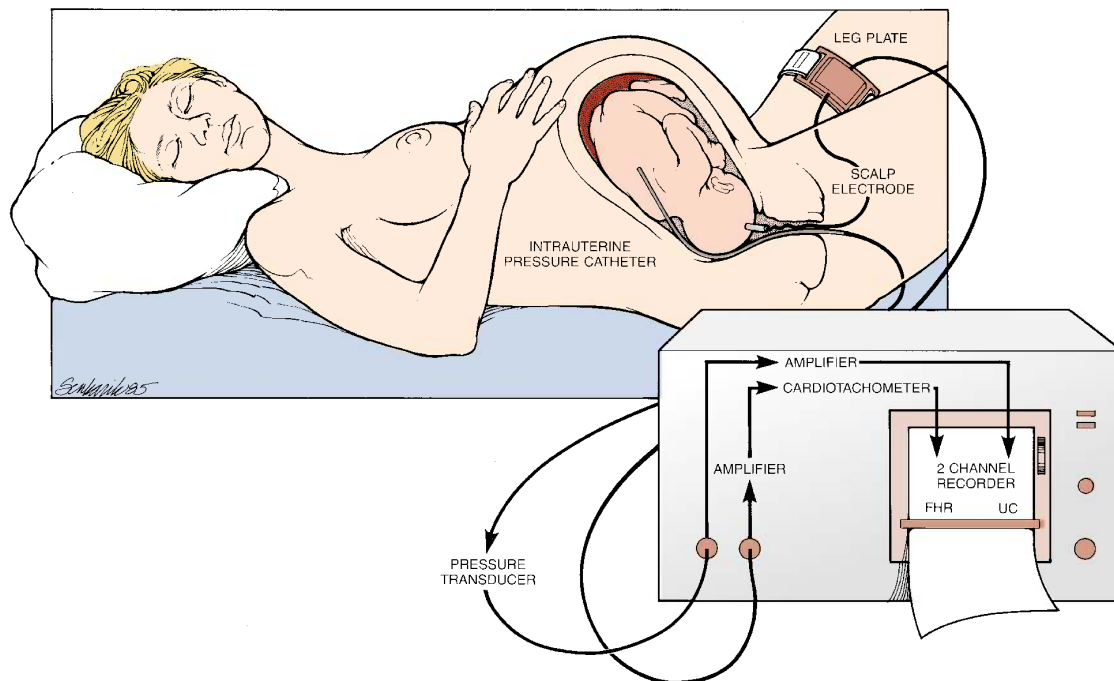


Fig. 17.3 Techniques Used for the Direct Monitoring of Fetal Heart Rate (FHR) and Uterine Contractions (UC). The fetal electrocardiogram is obtained by direct application of the scalp electrode, which is then attached to a leg plate on the mother's thigh. The signal is transmitted to the monitor, where it is amplified, counted by the cardi tachometer, and then recorded. Uterine contractions are assessed with an intrauterine pressure catheter connected to a pressure transducer. This signal is then amplified and recorded.

Association of Women's Health, Obstetric, and Neonatal Nurses (AWHONN); other relevant stakeholders; and scientists in the field to discuss nomenclature for EFM patterns, as well as the state of the science.³⁵ Historically, despite the fact that EFM was widely used in clinical practice and interpreted at the bedside by clinicians and nurses alike, the nomenclature and documentation were different, creating significant confusion. At the 2008 consensus conference, it was acknowledged that a single common nomenclature endorsed by all parties and used at the bedside was critical. It was also acknowledged that the prior two-tiered designation of EFM patterns as either "reassuring" or "nonreassuring" was inadequate and did not fully capture the range of patterns seen at the bedside or reflect the existing evidence of association. Therefore a new three-tiered nomenclature system was designated and endorsed by all of the stakeholders shortly thereafter. Since then all EFM patterns have been designated by the category system as category I (normal), category II (indeterminate), or category III (abnormal) (Table 17.2).³⁶

EFM patterns are interpreted clinically with visual observation in a stepwise approach. Several observed components are then

used to make an overall category designation. The first step is determination of the baseline heart rate, which can be defined by at least 2 minutes in any given 10-minute period. A normal range is 110 to 160 beats/min. A baseline heart rate greater than 160 beats/min defines tachycardia, and one below 110 beats/min defines bradycardia. Next, at the baseline, the magnitude of variability is determined, which is the amplitude of the oscillation around the baseline. There are four variability definitions: absent (no oscillation) (Fig. 17.4), minimal (Fig. 17.5), moderate (6 to 25 beats/min) (Fig. 17.6), and marked (>25 beats/min) (Fig. 17.7). A specific pattern of repetitive oscillation is referred to as sinusoidal (Fig. 17.8). Though rare, it is important to identify as it has been associated with fetal pathology such as anemia.

Accelerations are rises in the FHR above the baseline of at least 15 beats/min that last at least 15 seconds before returning to the baseline (Fig. 17.9). At less than 32^{0/7} weeks, accelerations are defined less stringently, as an increase of at least 10 beats/min that lasts at least 10 seconds before returning to the baseline. Physiologically, accelerations are thought to be due to fetal movement.

TABLE 17.2 Standard Fetal Heart Rate (FHR) Definitions

Pattern	Definition
Baseline	Mean FHR rounded to increments of 5 beats/min in a 10-min window excluding accelerations, decelerations, and periods of marked FHR variability (>25 beats/min). There must be at least 2 min of identifiable baseline segments (not necessarily contiguous) in any 10-min window or the baseline for that period is indeterminate. • Normal baseline FHR range: 110–160 beats/min • <i>Tachycardia</i> is defined as an FHR baseline >160 beats/min • <i>Bradycardia</i> is defined as an FHR baseline <110 beats/min
Variability	Fluctuations in the FHR baseline are irregular in amplitude and frequency and are visually quantitated as the amplitude of the peak to the trough in beats per minute. • Absent: amplitude range undetectable • Minimal: amplitude range detectable but ≤5 beats/min • Moderate (normal): amplitude range 6–25 beats/min • Marked: amplitude range >25 beats/min
Accelerations	Abrupt increase (onset to peak <30 s) in the FHR from the most recently calculated baseline. At ≥32 weeks, an acceleration peaks ≥15 beats/min above baseline and lasts ≥15 s but <2 min. At <32 weeks, acceleration peaks ≥10 beats/min above baseline and lasts ≥10 s but <2 min. Prolonged acceleration lasts ≥2 min but <10 min. Acceleration ≥10 min is a baseline change.
Early	A gradual (onset to nadir ≥30 s) decrease in FHR during a uterine contraction. Onset, nadir, and recovery of the deceleration occur at the same time as the beginning, peak, and end of the contraction, respectively.
Late	A decrease in FHR is gradual (onset to nadir ≥30 s) during a uterine contraction. Onset, nadir, and recovery of the deceleration occur after the beginning, peak, and end of the contraction, respectively.
Variable	Decrease in the FHR is abrupt (onset to nadir <30 s); it is ≥15 beats/min below the baseline and lasts ≥15 s but <2 min.
Prolonged	Deceleration is ≥15 beats/min below baseline and lasts ≥2 min or more but <10 min. Deceleration ≥10 min is a baseline change.
Sinusoidal pattern	Pattern in FHR baseline is smooth, sine wave–like, and undulating with a cycle frequency of 3–5/min that persists for at least 20 min.
Categories	
Category I	Includes all the following: • Baseline: 110–160 bpm • Moderate baseline variability • Variable and late decelerations absent • Presence or absence of early decelerations • Presence or absence of accelerations
Category II	All fetal heart rate tracings not classified as Category I or Category III.
Category III	Includes either: • Absent baseline variability with recurrent late decelerations, recurrent variable decelerations, and/or bradycardia • Sinusoidal pattern

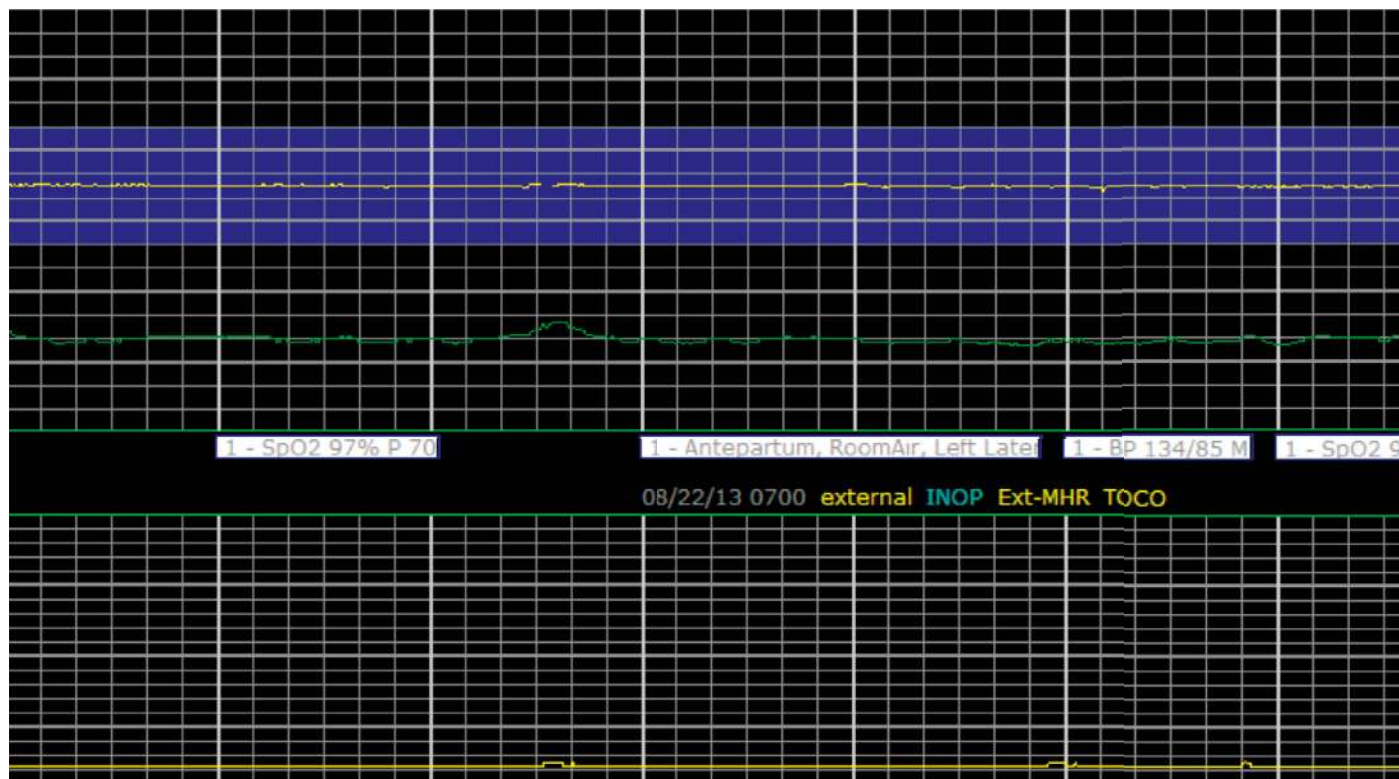


Fig. 17.4 Absent Variability.

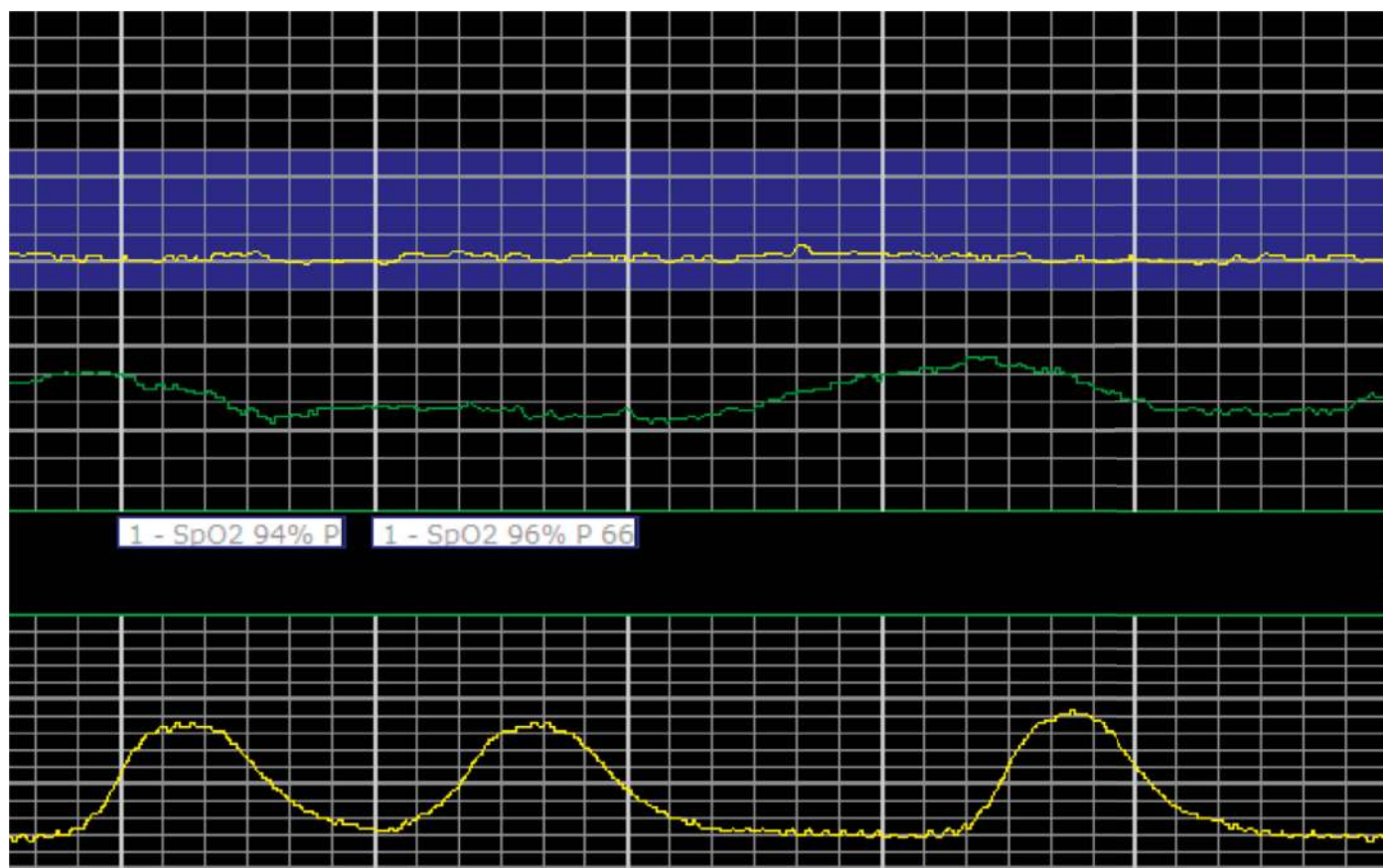


Fig. 17.5 Minimal Variability.

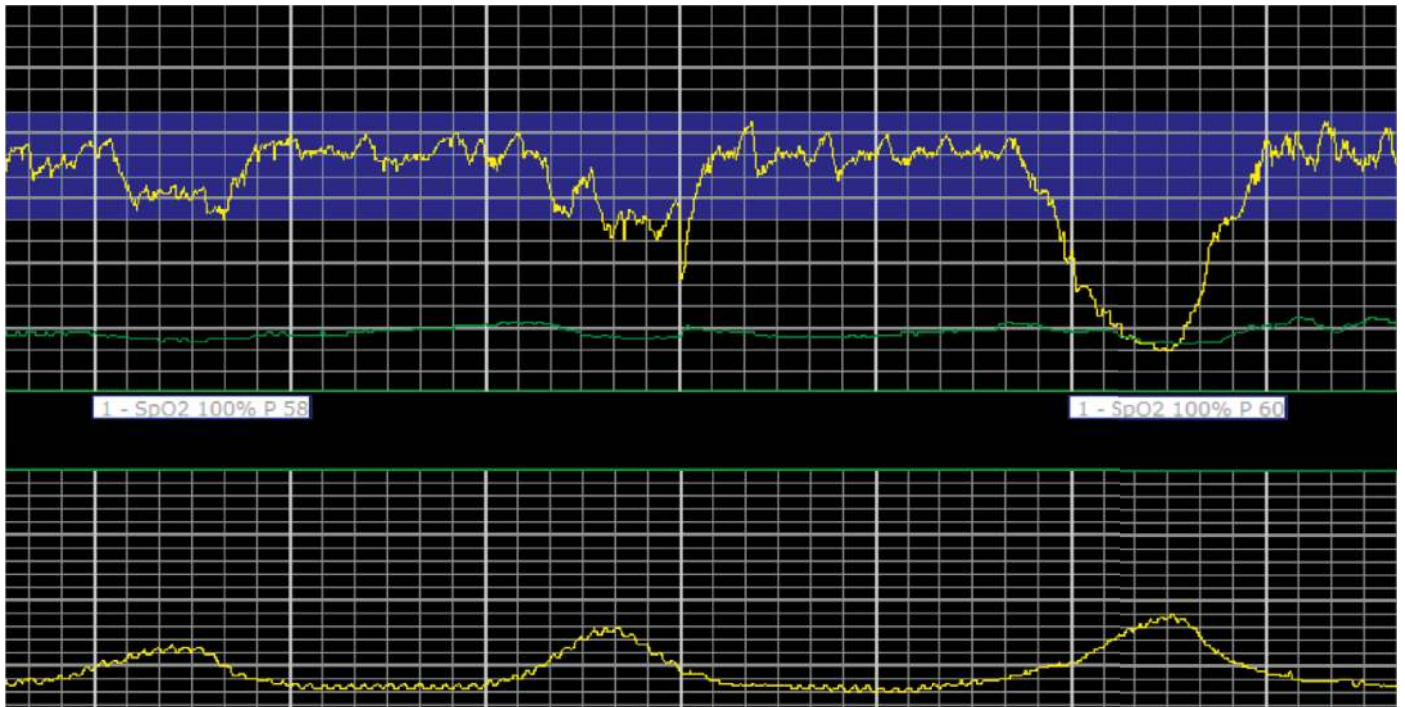


Fig. 17.6 Moderate Variability.

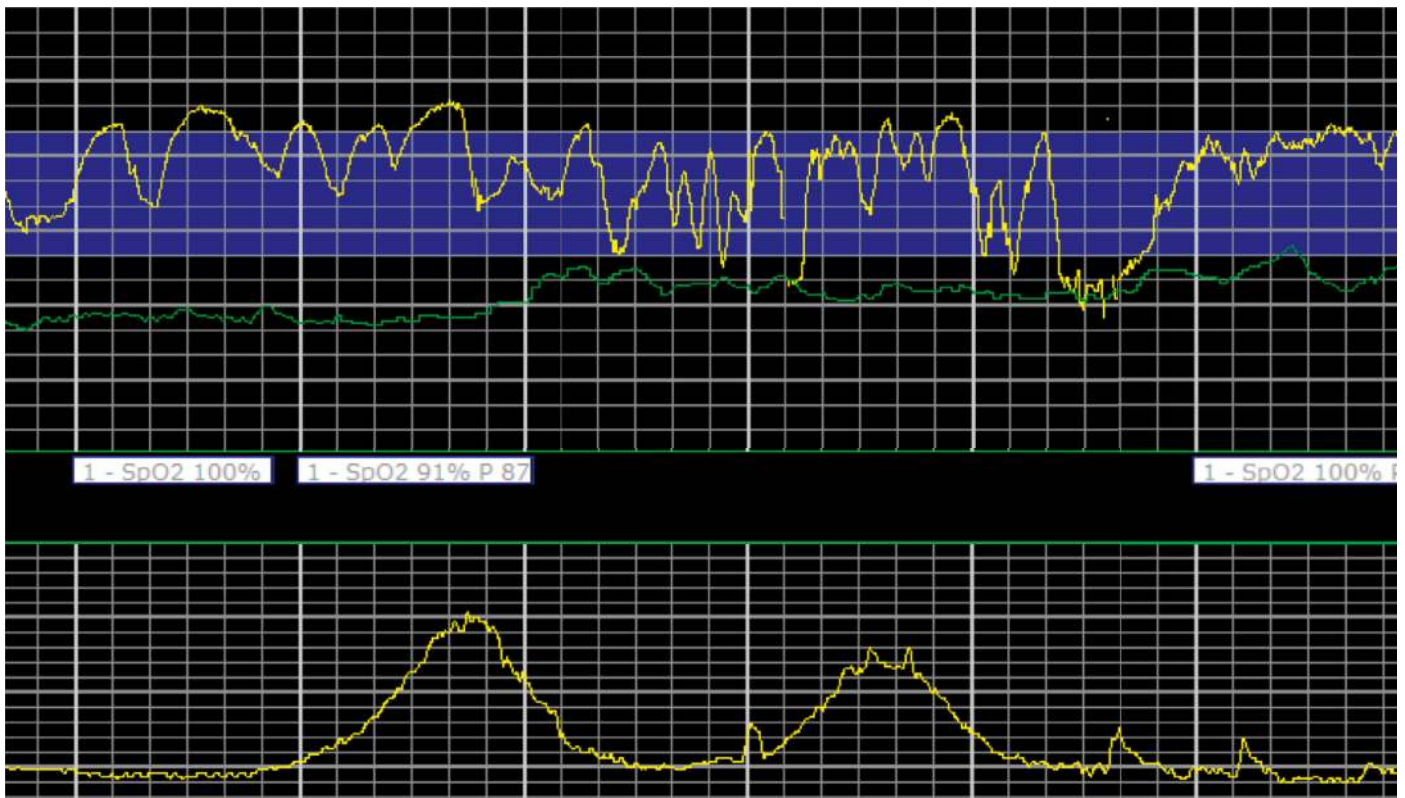


Fig. 17.7 Marked Variability.

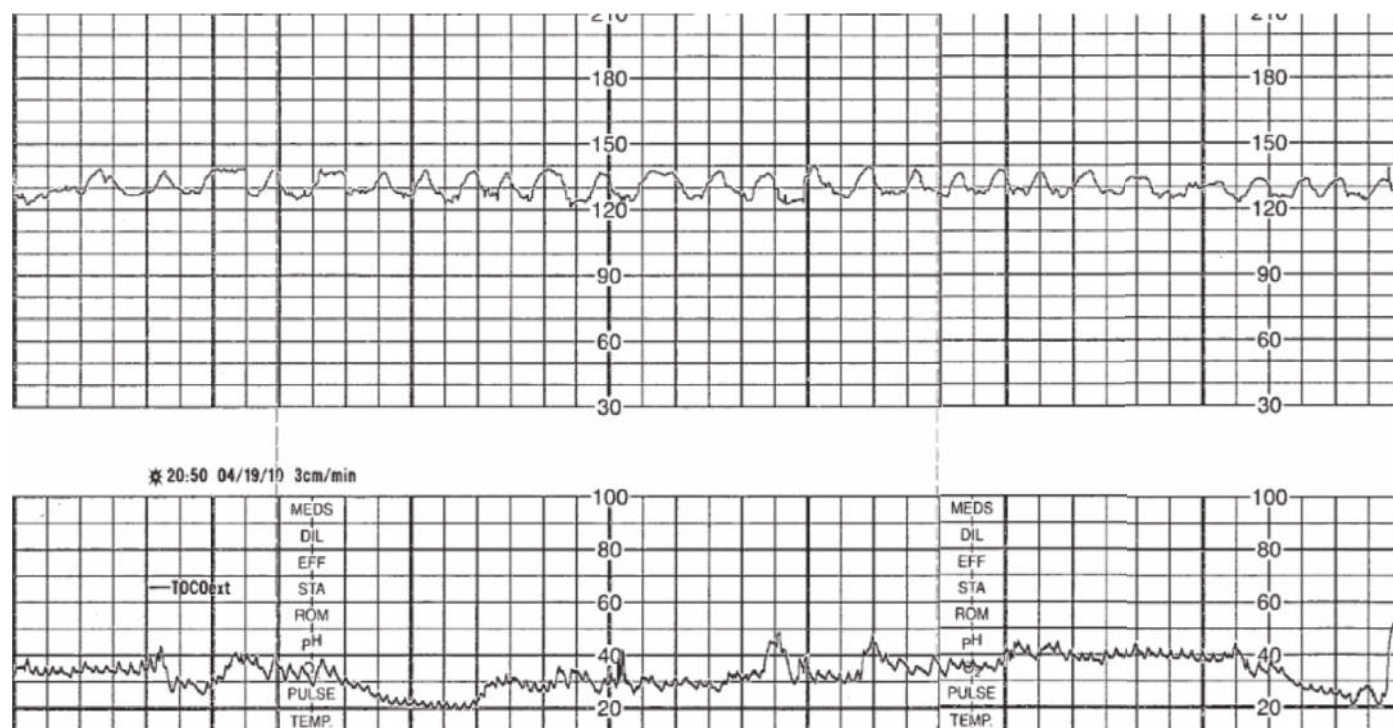


Fig. 17.8 Sinusoidal pattern.

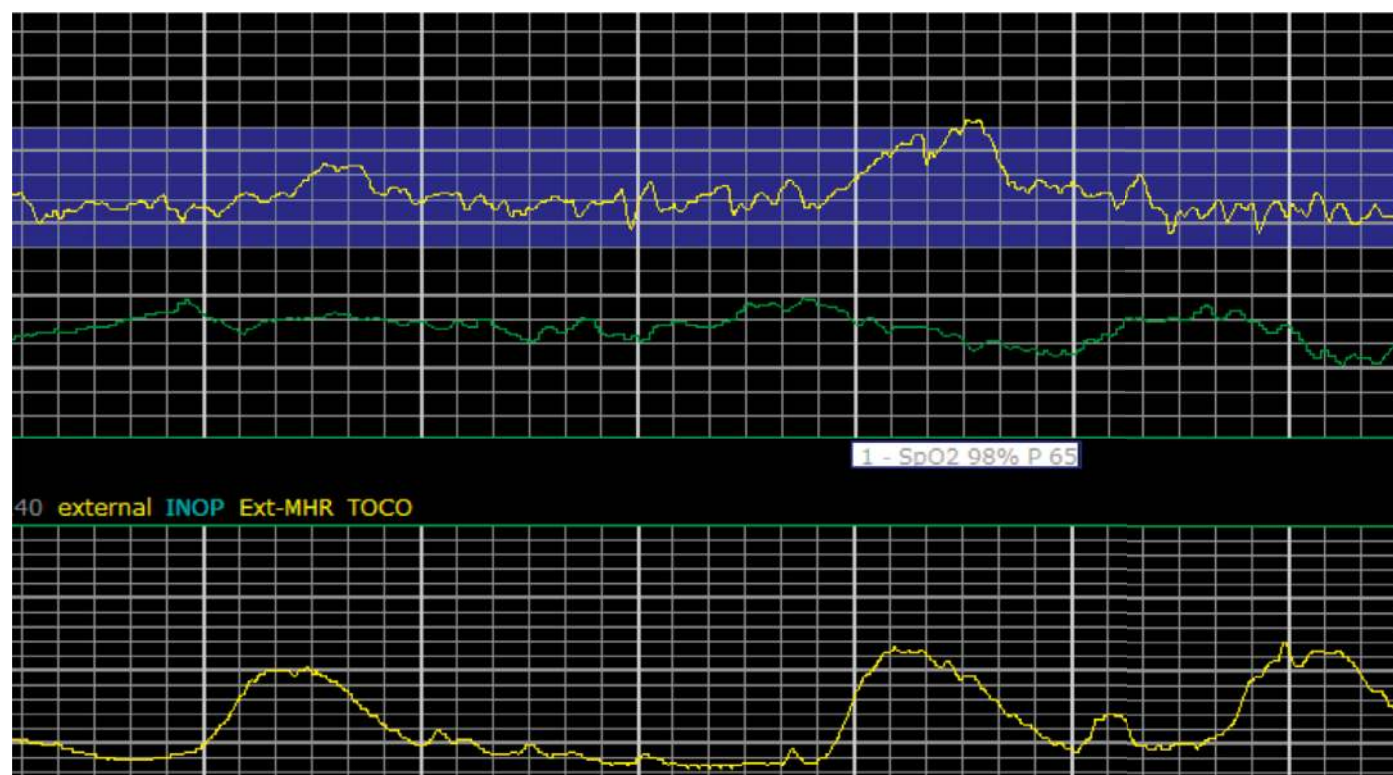


Fig. 17.9 Category I With Acceleration.

It is important to determine the presence or absence of decelerations. Decelerations are decreases in the FHR below the baseline that then return to baseline. **There are four different types of decelerations. Because their temporal relationship to contractions helps to define the deceleration type for some, accurate assessment of contraction timing is prudent.** Early decelerations are marked by a gradual decrease followed by a gradual return to baseline, which is concluded by the time the contraction ends; it lasts less than 2 minutes and generally is a mirror image of the contraction in terms of onset, nadir, and resolution. Early decelerations are thought to be a physiologic response to fetal head compression and thus are seen most commonly during active periods of fetal descent in labor.

Variable decelerations are characterized by an abrupt descent and an abrupt return to baseline of at least 15 beats/min lasting less than 2 minutes (Fig. 17.10). The nadir of the deceleration is reached in less than 30 seconds, and the shape is often described as a “V” or “U.” Variable decelerations are caused by umbilical cord compression, which can happen at any time due to the relative movements and positions of mother and fetus and particularly during labor contractions and the cardinal movements. When the umbilical cord is compressed, the thin-walled umbilical vein is typically compressed first, which reduces fetal preload and can produce a short, transient compensatory increase in FHR that immediately precedes the variable deceleration. When the umbilical arteries are compressed as the cord is fully occluded, baroreflexes and chemoreflexes mediate a decrease in heart rate. As compression of the umbilical cord resolves, the thick-walled arteries regain flow first, followed by the vein, often creating the symmetric appearance of variable decelerations (Fig. 17.11).

Late decelerations are characterized by a slow descent and slow return to baseline (Fig. 17.12). The timing of the nadir of late decelerations is after the peak of the contraction, and the deceleration resolves after the contraction has concluded. Late decelerations are thought to be mediated by baroreflexes and chemoreceptor responses.

Late decelerations are generally associated with fetal hypoxia but not necessarily acidemia (Fig. 17.13).

Prolonged decelerations are characterized not by shape or timing relative to contractions but by duration (Fig. 17.14). Any deceleration of at least 15 beats/min regardless of shape that lasts more than 2 minutes but less than 10 minutes is a prolonged deceleration. Decelerations that occur with more than 50% of contractions during any given period are considered recurrent, regardless of type.

The combination of features over any 10-minute time interval are considered together and a category is assigned to inform communication, documentation, and management (Table 17.2). Category I is considered a normal EFM pattern (Figs. 17.15 and 17.16), whereas category III is abnormal (Figs. 17.17 to 17.19). The majority of patterns seen during labor are category II, encompassing a wide spectrum of patterns (Figs. 17.20 to 17.27).³⁷

Terms No Longer Used

A few historically described EFM features were not included in the 2008 nomenclature system because—although they were once thought to be clinically informative—no evidence has demonstrated an association with those patterns and clinically meaningful outcomes. In terms of variability, historically there was a distinction between short-term and long-term variability. These terms attempted to distinguish between beat-to-beat variation at the baseline (short-term variability) and periodic variation around the baseline (long-term variability). However, despite the hypothesis that decreased short-term variability was associated with acidemia, larger studies of higher quality did not demonstrate any association with meaningful clinical outcomes, including acidemia. Thus this distinction is no longer used clinically and accordingly not included in the clinical nomenclature standard set forth by NICHD.³⁶ Similarly, it was historically thought that the variability within a deceleration, particularly at the nadir, had an association with acidemia. However, no studies have provided evidence

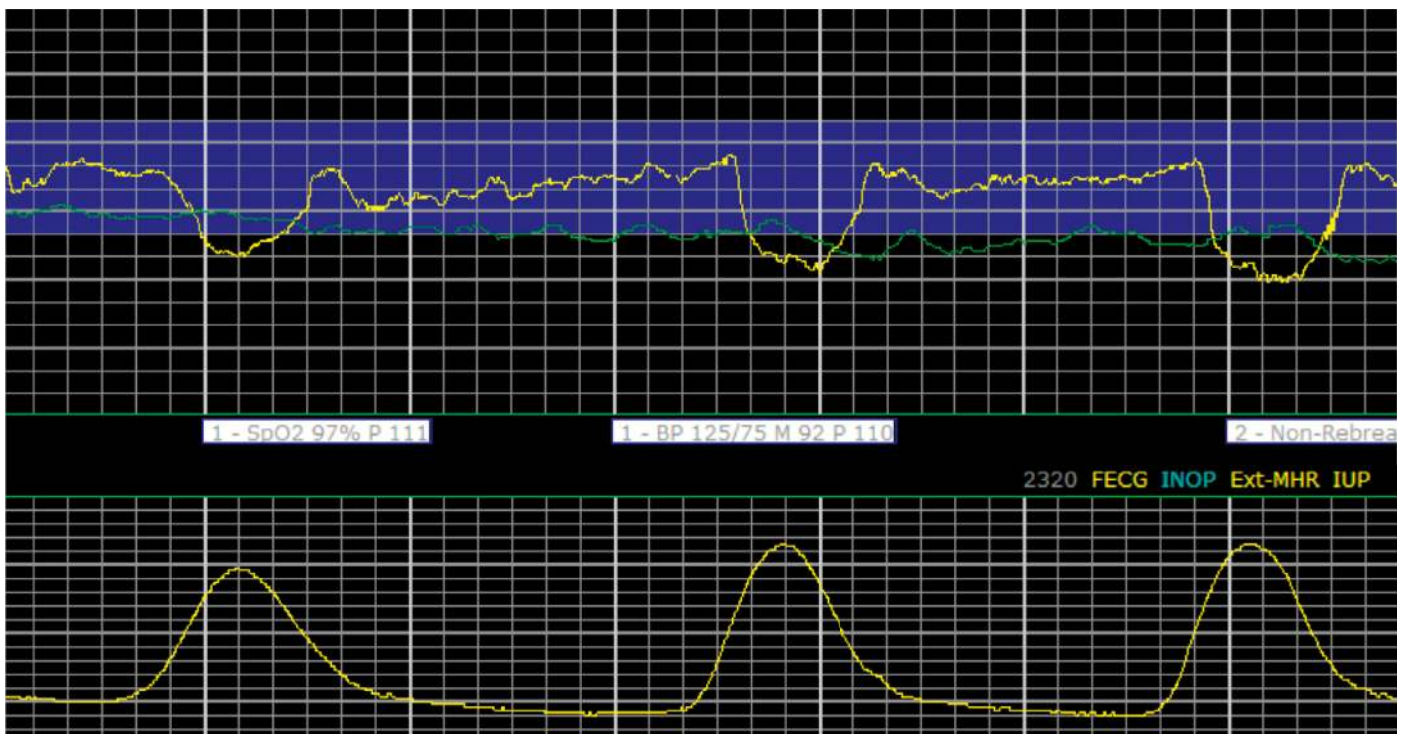


Fig. 17.10 Variable Decelerations.

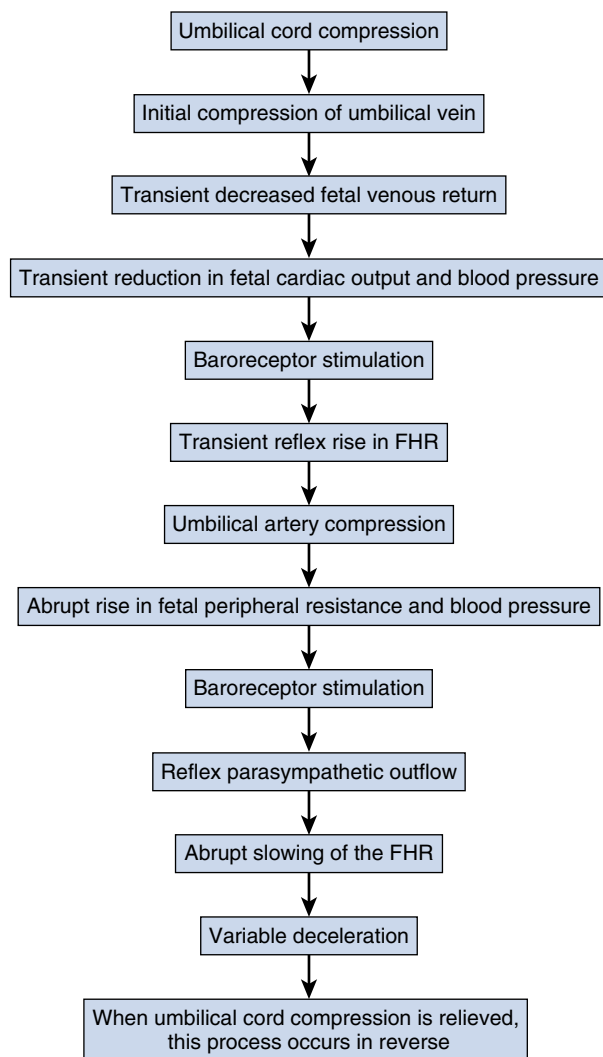


Fig. 17.11 Mechanism of a Variable Deceleration. (FHR, Fetal heart rate.)



Fig. 17.12 Late Deceleration.

for that, and the current guidelines for determining variability say to do so when the fetus is at the baseline heart rate.

Two features of decelerations were also historically thought to be important for clinical interpretation. The term *shoulder* referred to the

brief increase in FHR often seen preceding and/or after the deceleration. The term *overshoot* referred to the FHR following a deceleration, which would rise above the established baseline before returning to it. All of these features were thought to be clinically informative.³⁸ However, they have never been shown to be associated with acidemia and/or adverse neonatal outcomes³⁹ and thus were appropriately removed from the nomenclature system.³⁶ These descriptions should not be included in clinical communication or documentation unless new evidence emerges and recommendations change.

UTERINE ACTIVITY

Uterine monitoring is an important part of intrapartum fetal monitoring because it enables the interpretation of EFM patterns⁴⁰ in the context of uterine activity. There are two main reasons why specific uterine activity is important apart from interpreting fetal well-being in the context of labor progress. First, the FHR decelerations should be interpreted timed with uterine contractions to establish whether or not late decelerations are occurring. Decelerations that nadir after the apex of the contractions and resolve after the resolution of the contractions are late decelerations that require clinical evaluation. Second, tachysystole is an important consideration because it has been suggested that decreased oxygenation during excessive contractions could lead to fetal hypoxia over time.

Tachysystole is defined as greater than 5 contractions in a 10-minute period averaged over 30 minutes. In the presence of tachysystole, the accompanying FHR pattern is important to evaluate. A prospective study of 584 women at term undergoing misoprostol labor induction examined the association between tachysystole in the first 4 hours after the initial dose of misoprostol and adverse infant outcomes. It found that tachysystole was associated with fetal decelerations but that there was no association with neonatal morbidity.⁴¹ In contrast, a retrospective cohort study of more than 50,000 deliveries examined the association between tachysystole and neonatal morbidity. It found that tachysystole was associated with adverse neonatal outcomes, specifically the number of episodes of tachysystole and composite neonatal morbidity (RR, 1.15; 95% CI, 1.07 to 1.23).⁴² It identified risk factors for tachysystole that included oxytocin or misoprostol use and epidural anesthesia. Because tachysystole may be associated with adverse infant outcomes, NICHD nomenclature recommends making a clinical distinction between tachysystole with (Fig. 17.28) or without FHR abnormalities (Fig. 17.29).⁴³ **When tachysystole occurs during spontaneous labor without FHR abnormalities, no intervention is needed and continued observation is appropriate. But when category II or III EFM patterns occur in the setting of tachysystole, measures should be taken to resolve the tachysystole in order to resolve the EFM patterns.** It is also reasonable to make efforts to resolve tachysystole that occurs in the setting of labor induction or augmentation whenever possible to prevent FHR abnormalities from developing.

OTHER NOMENCLATURE

Other nomenclature systems are used outside the United States. For example, the Federation for Gynecology has its own nomenclature system, as does the National Institute for Health and Care Excellence, although both share many similarities with the category system used in the United States.⁴⁴ A study comparing the interobserver variability in using the three different systems to identify pathologic EFM patterns in labor found them to have a high level of agreement.⁴⁵

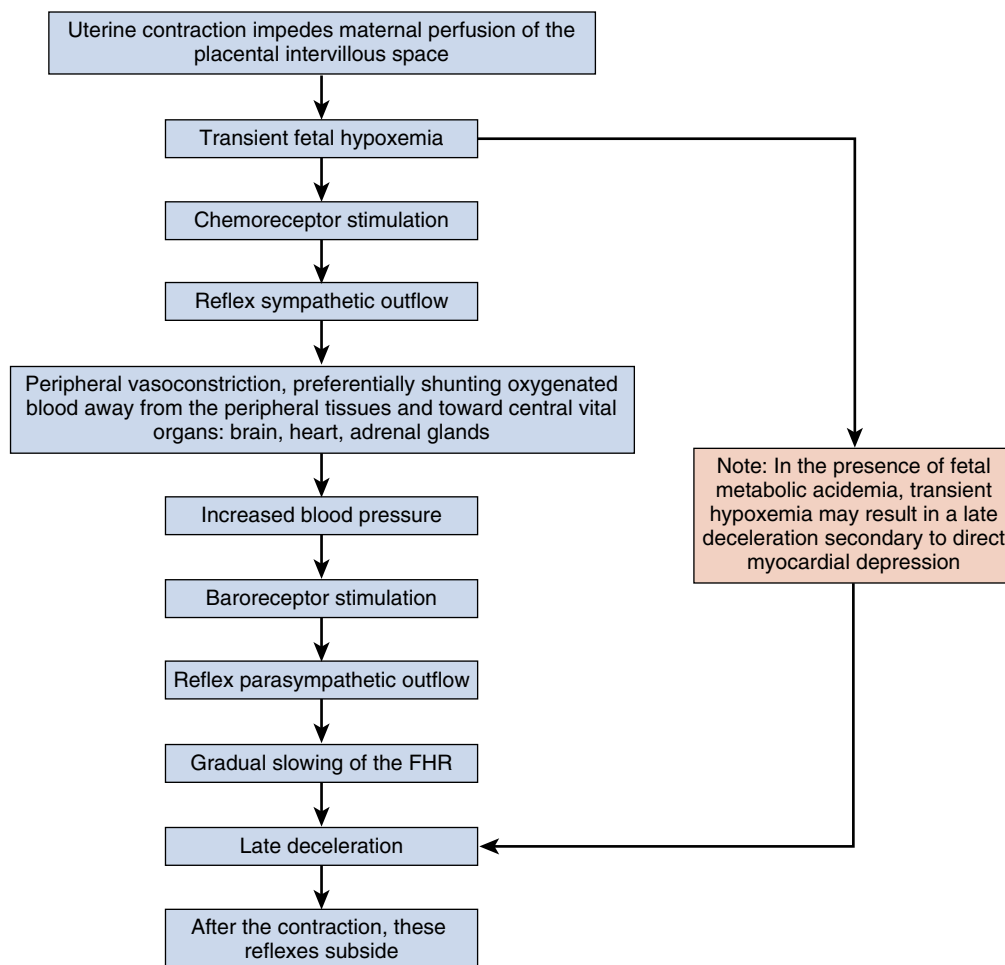


Fig. 17.13 Mechanism of Late Deceleration. (FHR, Fetal heart rate.)

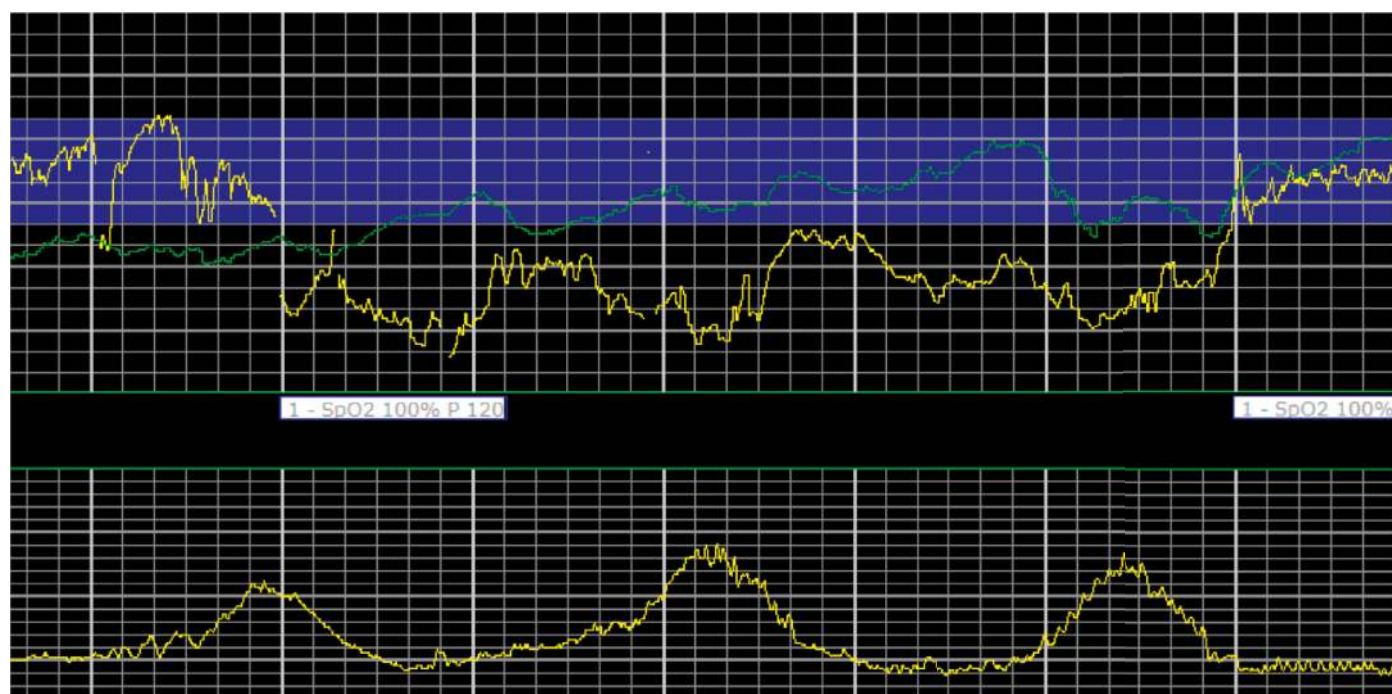


Fig. 17.14 Prolonged Deceleration.

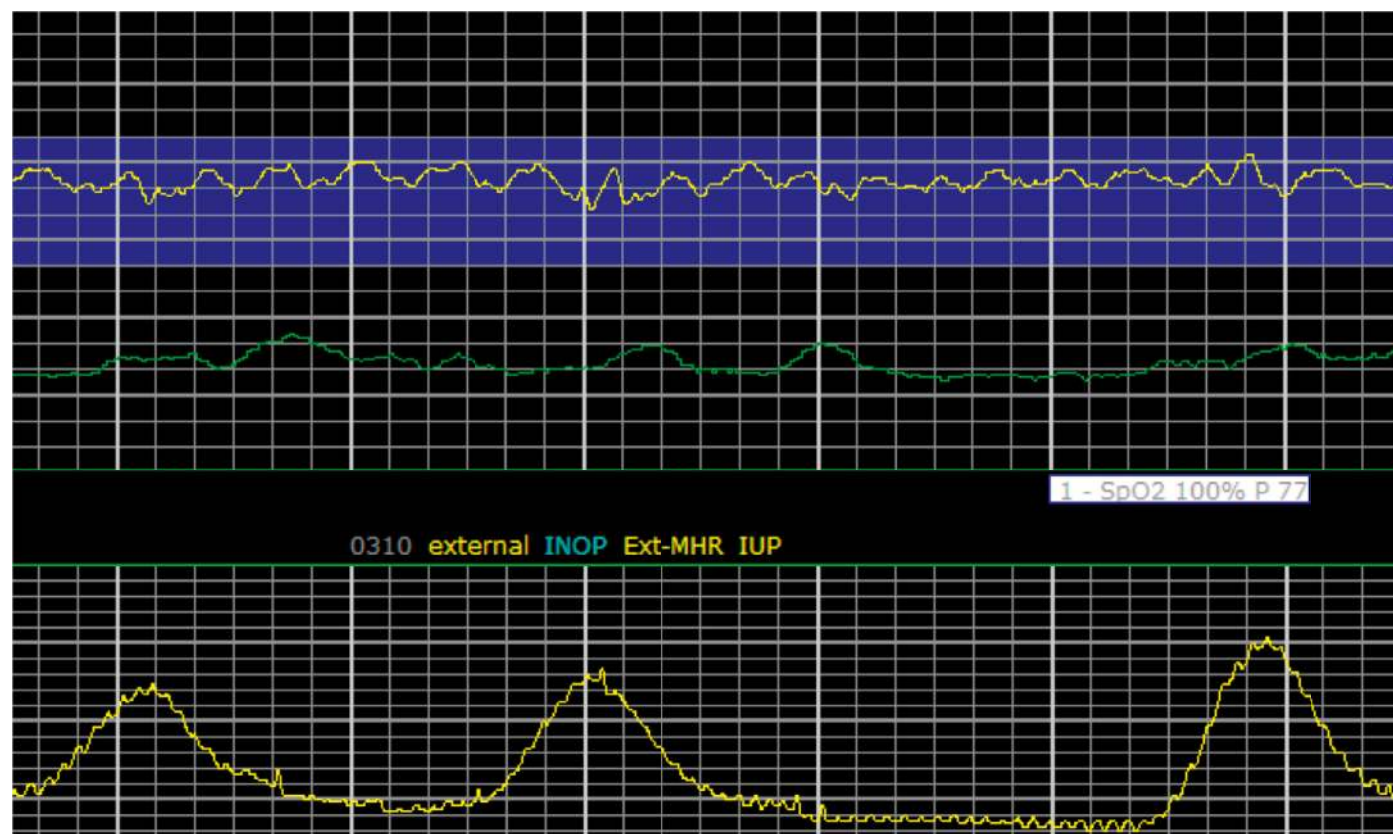


Fig. 17.15 Category I Without Accelerations.

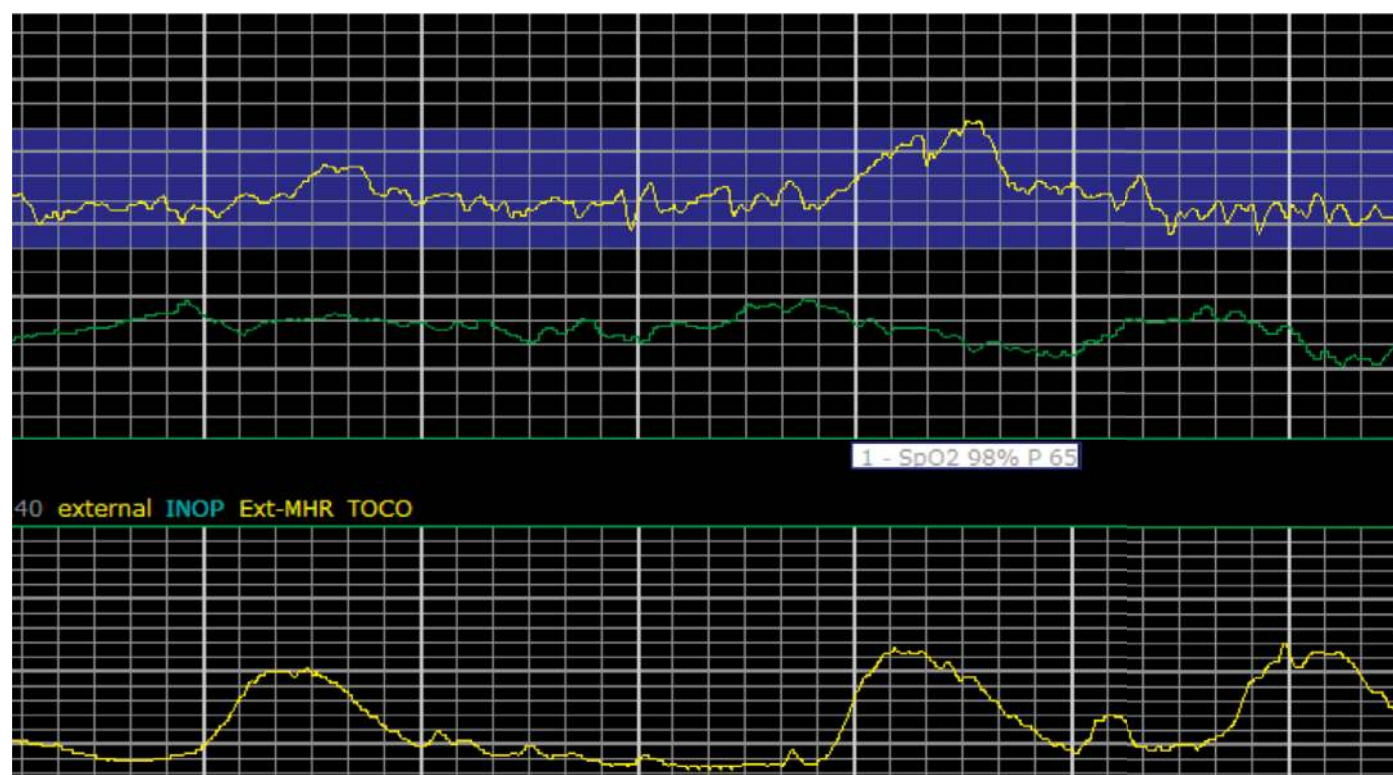


Fig. 17.16 Category I With Acceleration.

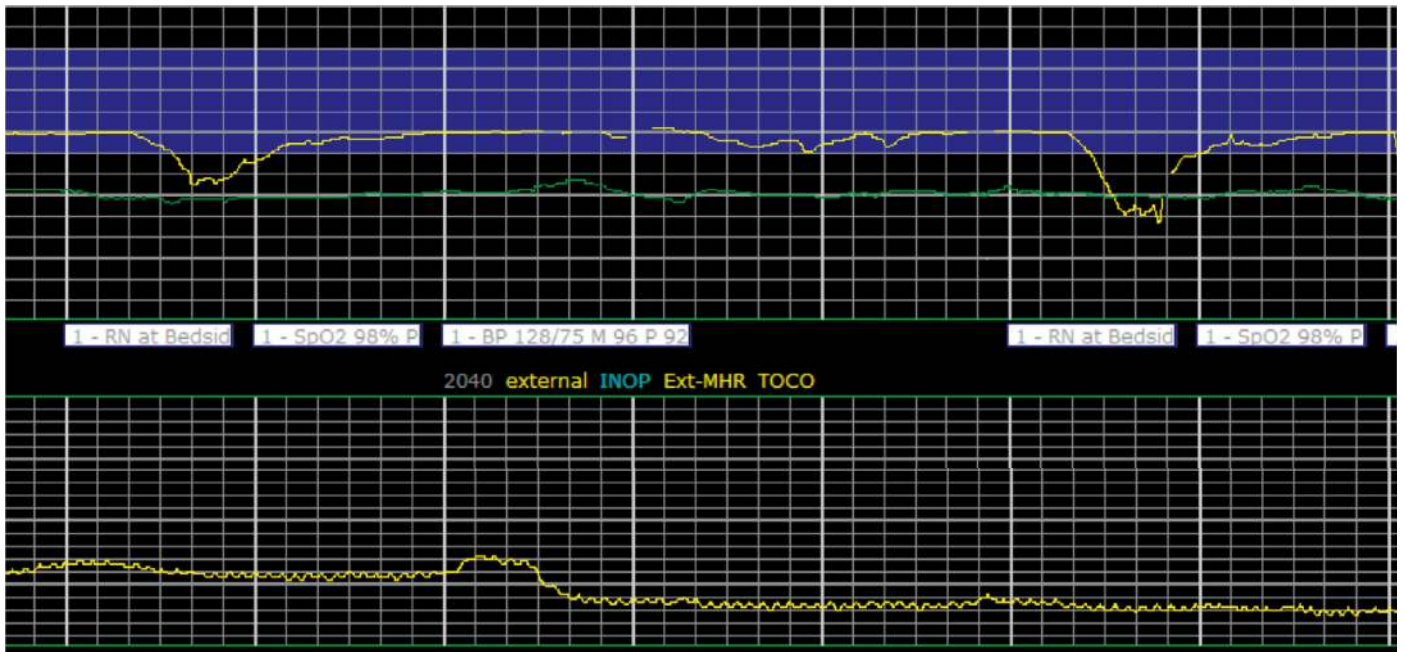


Fig. 17.17 Category III 1.

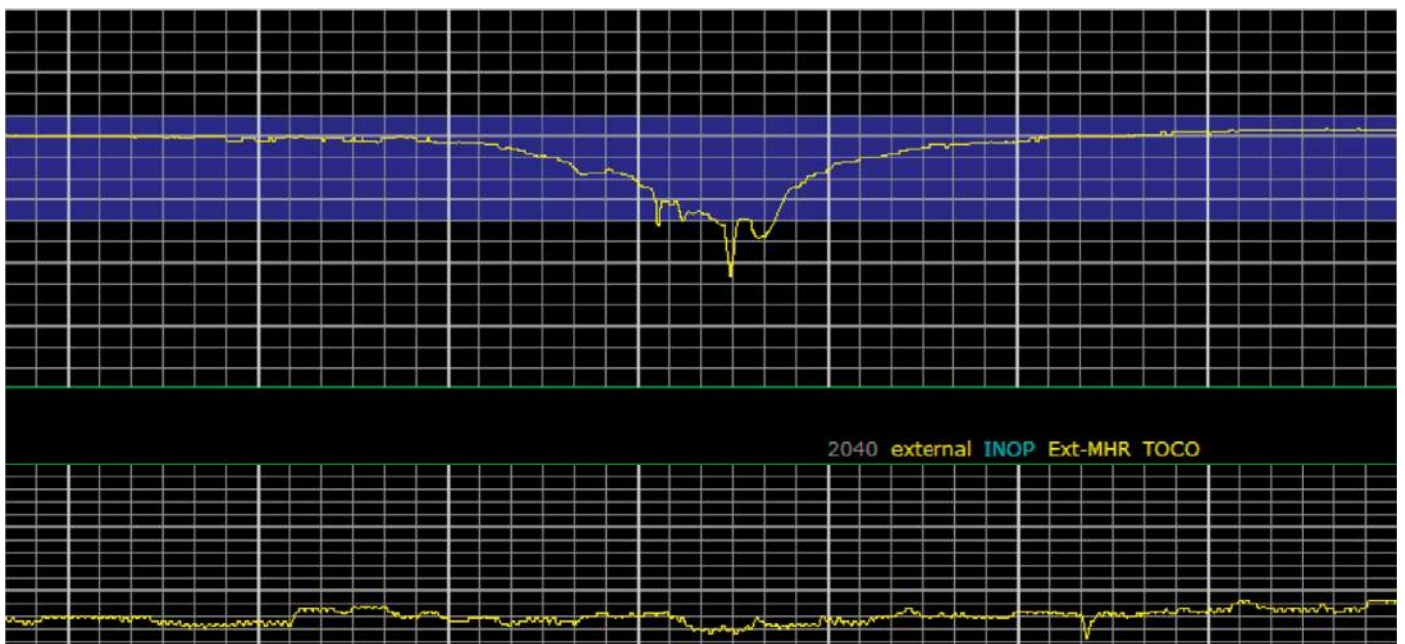


Fig. 17.18 Category III 2.

INTERMITTENT MONITORING

Intrapartum EFM is used in a continuous fashion from the onset of labor in the majority of clinical circumstances. However, ACOG continues to hold, largely because the data for the effectiveness of continuous EFM are inconclusive, that intermittent EFM during labor is a reasonable option

for some low-risk patients. If intermittent monitoring is elected in selected patients, a 1:1 nurse-to-patient ratio is required. ACOG further recommends that fetal monitoring be performed every 30 minutes in the active phase of labor and every 15 minutes during the second stage of labor, based on expert opinion when intermittent monitoring is employed for selected low-risk women.⁴⁰

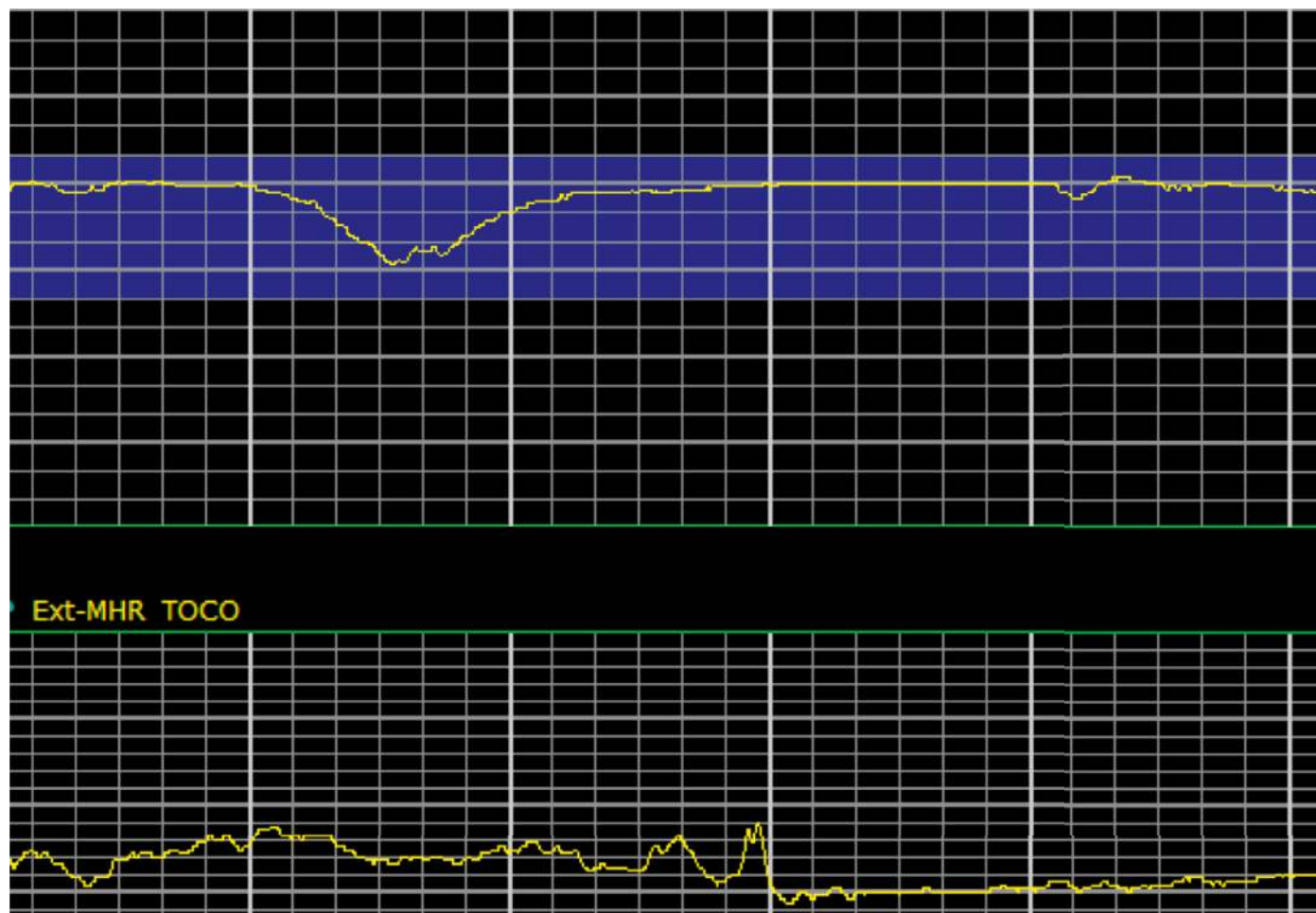


Fig. 17.19 Category III 3.



Fig. 17.20 Category II 1.

ELECTRONIC FETAL MONITORING PATTERNS ASSOCIATED WITH ACIDEMIA AND CLINICAL OUTCOMES

Several observational studies have demonstrated an association between specific EFM patterns in categories II and III and acidemia,

although the definitions of acidemia vary among studies. Early studies in selected populations demonstrated an association with tachycardia or bradycardia⁴⁴ and odds of a pH below 7.20, as well as absent variability and base deficit greater than 12 mmol/L.⁴⁵ Two additional studies demonstrated an association between absent or minimal variability, or late or prolonged decelerations, and a pH below 7.^{46,47} A cohort study of more than 10,000 women demonstrated that

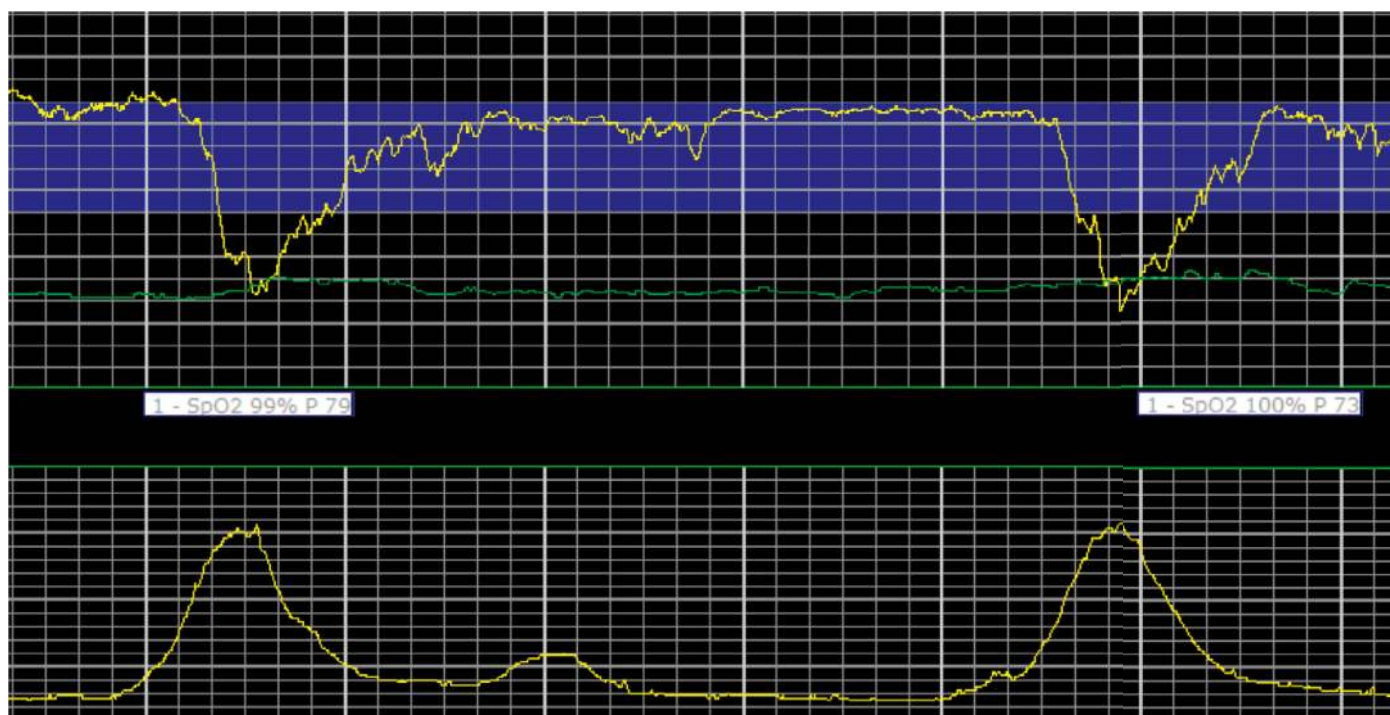


Fig. 17.21 Category II 2.

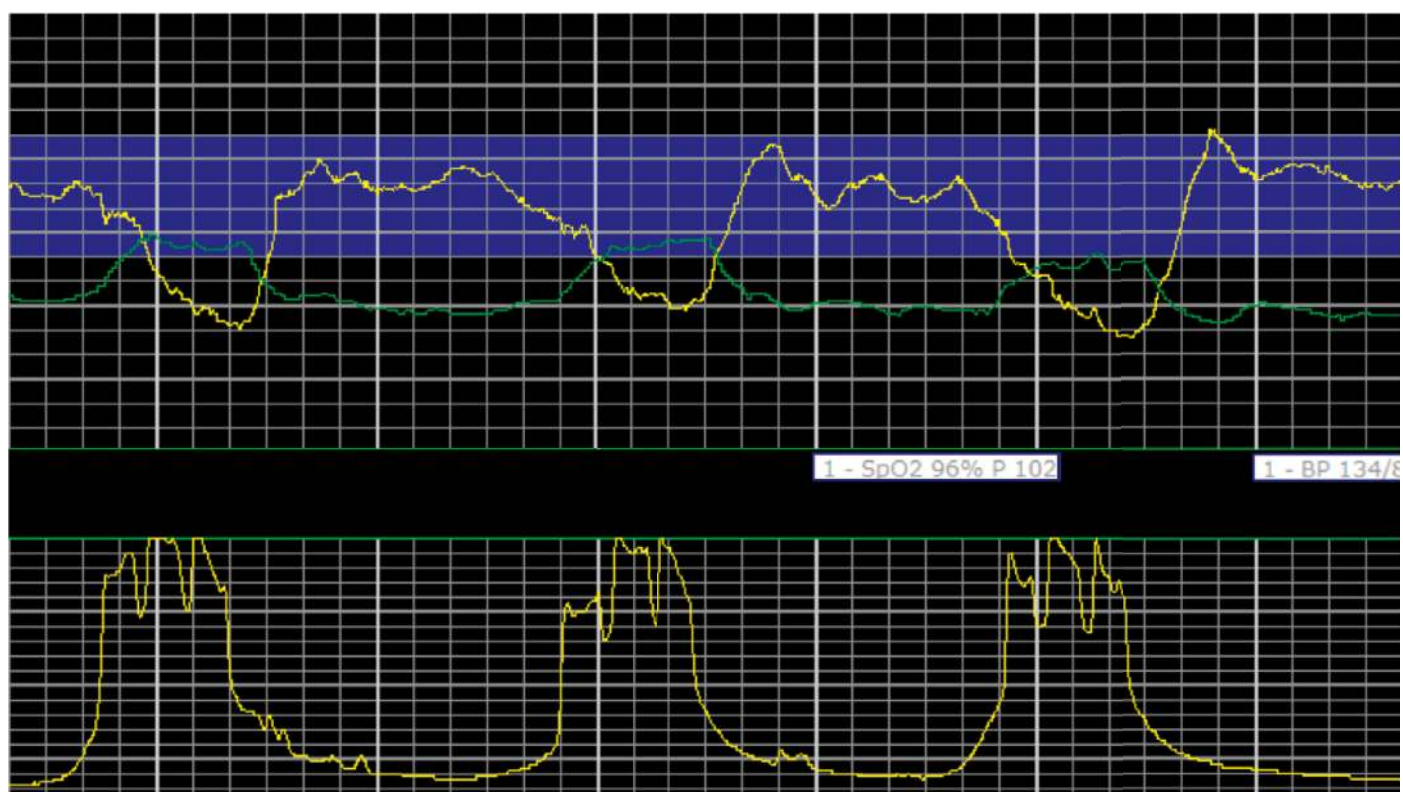


Fig. 17.22 Category II 3.

tachycardia, minimal or absent variability, or the presence of decelerations (variable, late, or prolonged) was associated with a pH below 7.10.^{48,49} A more contemporary cohort study of 5388 women at a tertiary center with a universal cord gas policy concurred with the prior work by

Sameshima, finding an association between tachycardia or decelerations (variable, late, or prolonged) and a pH below 7.10.³⁷ In a retrospective cohort study of more than 5000 women at term, terminal decelerations were not associated with an increased risk of acidemia or neonatal

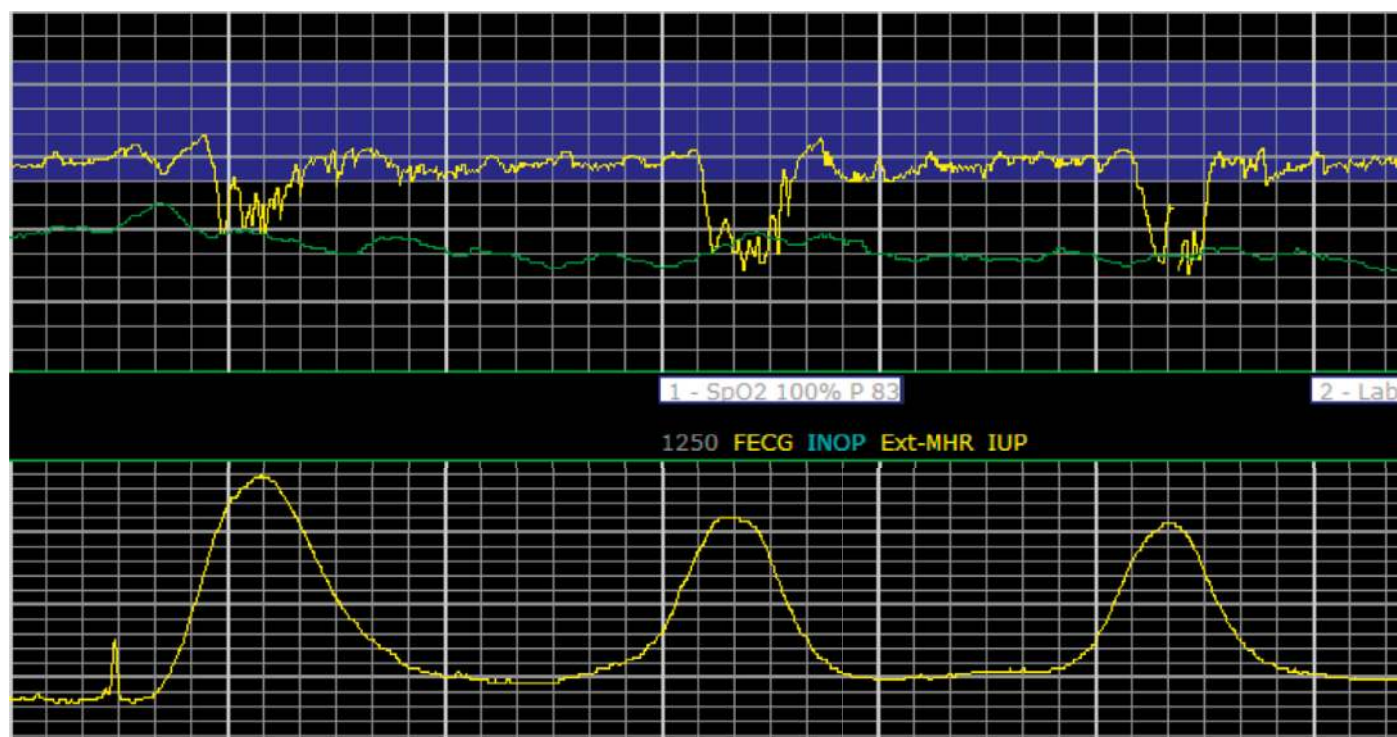


Fig. 17.23 Category II 4.

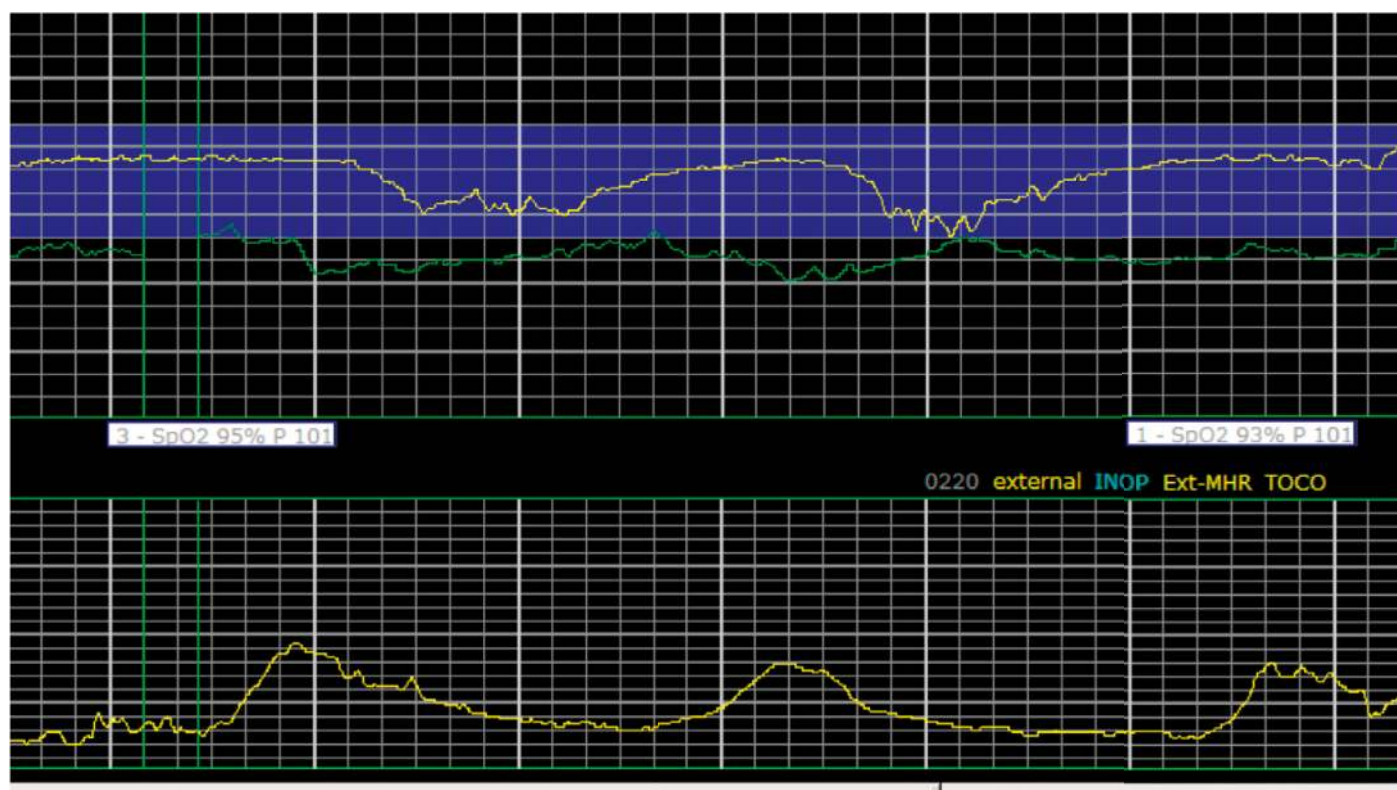


Fig. 17.24 Category II 5.

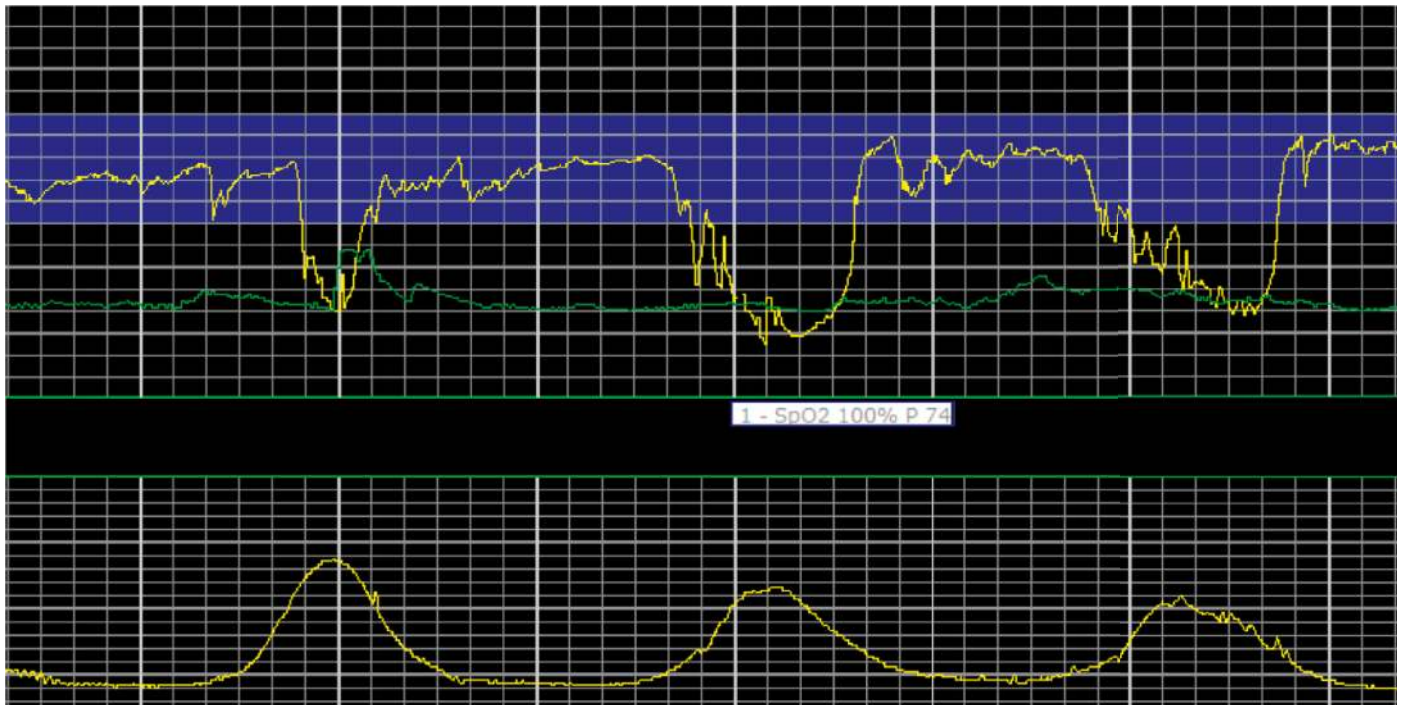


Fig. 17.25 Category II 6.

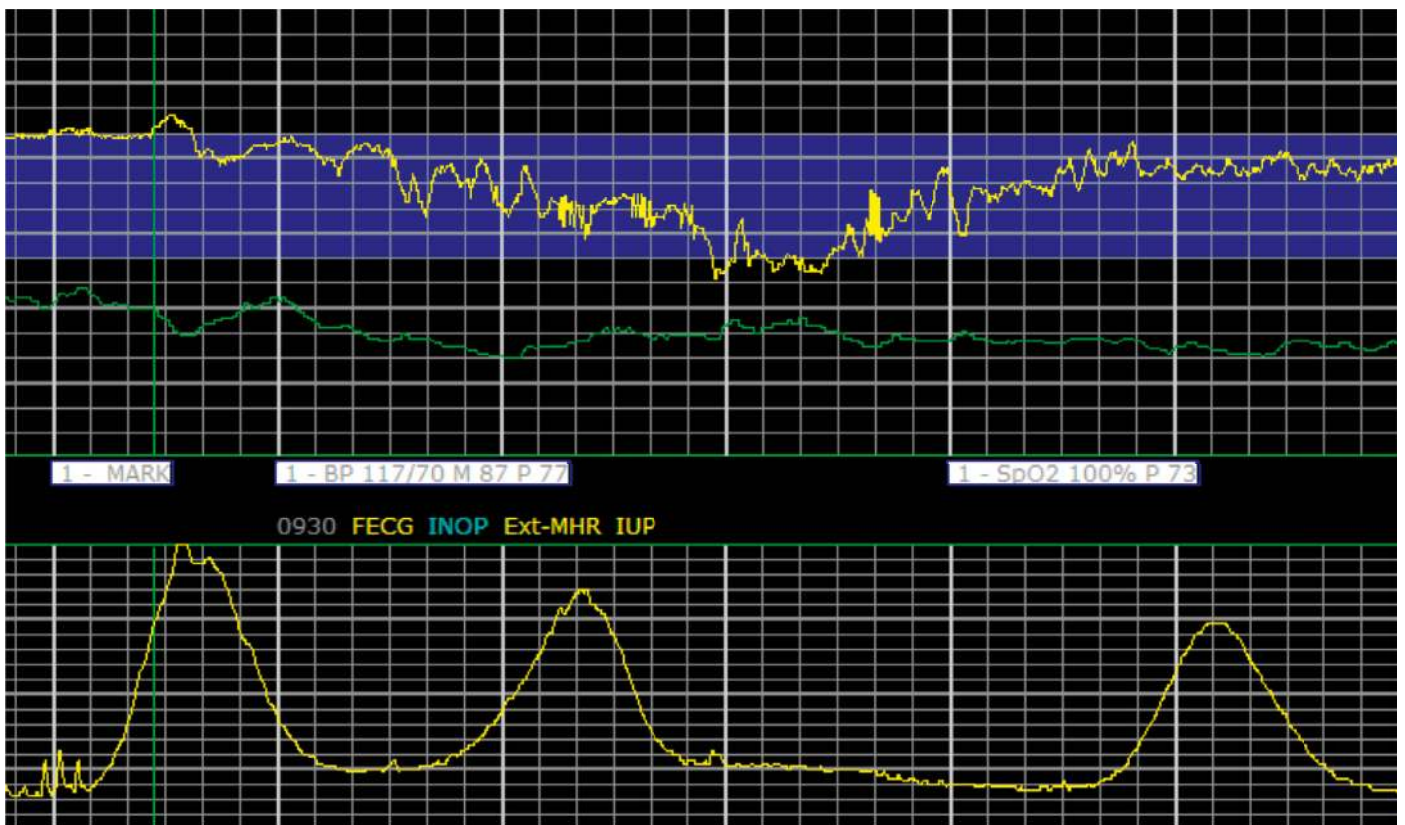


Fig. 17.26 Category II 7.

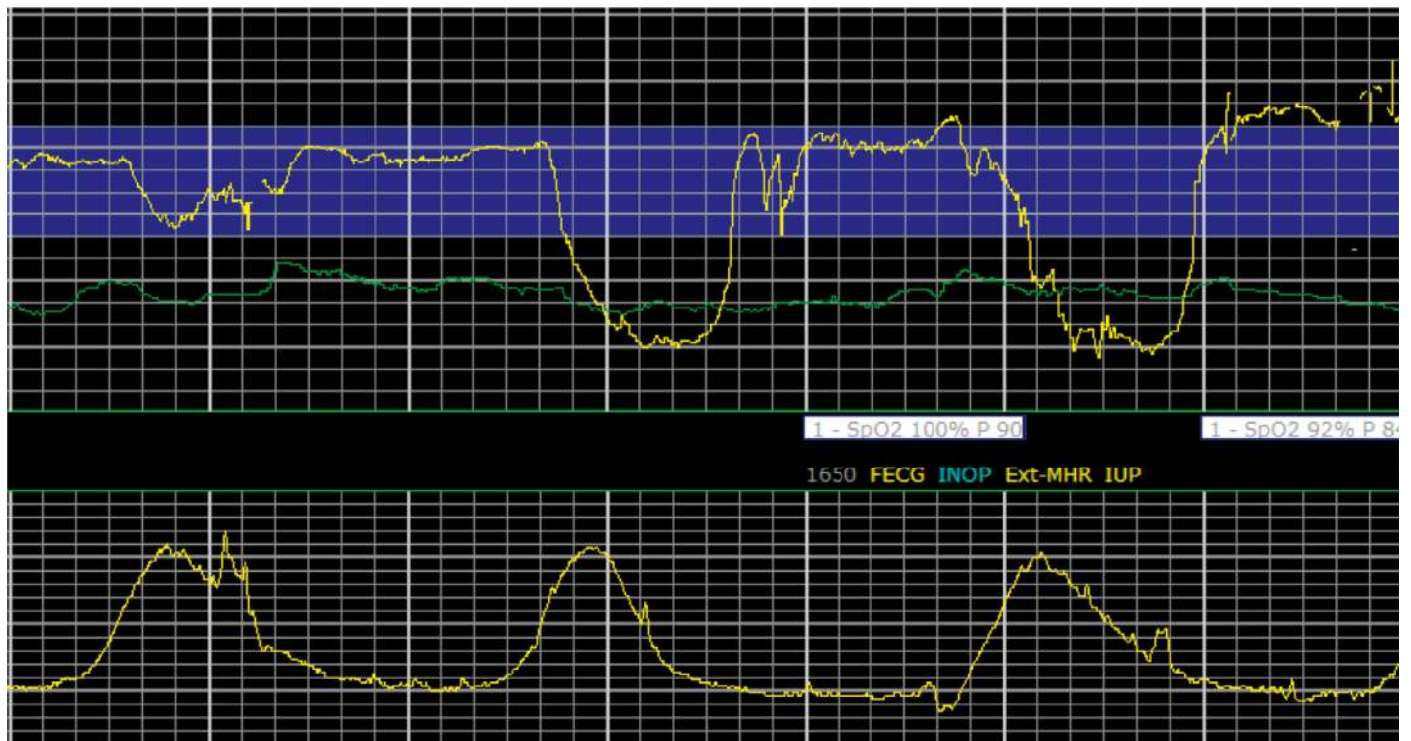


Fig. 17.27 Category II 8.

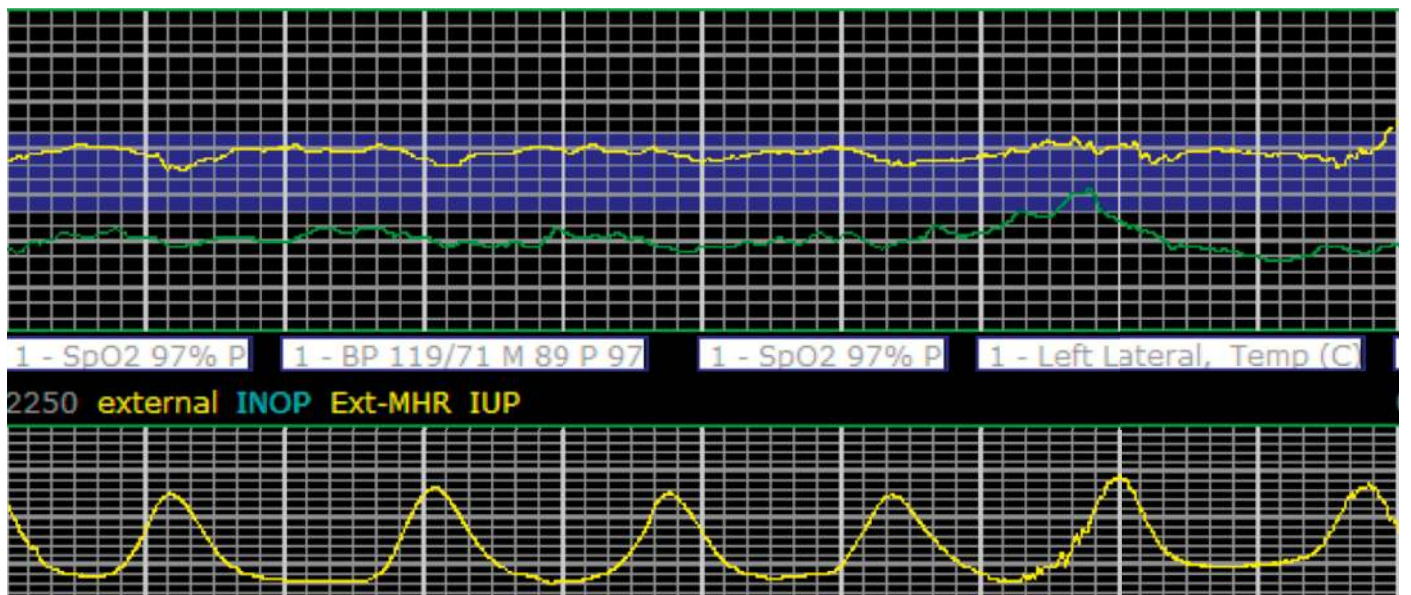


Fig. 17.28 Tachysystole With Abnormal Electronic Fetal Monitoring.

morbidity. However, among those with a terminal deceleration before delivery, bradycardia compared with a terminal deceleration lasting less than 10 minutes was associated with a significantly increased risk of acidemia.⁵⁰ A prospective cohort study of more than 8000 women at term demonstrated an association between any 10-minute period of category III and pH below 7.10 (aRR, 8.72; 95% CI, 2.92 to 26.04). It has been observed that some infants change baseline rate during labor; however, as long as the new baseline is within normal limits, there is no association between baseline change and acidemia regardless of

direction or magnitude of change.⁵¹ All of these studies support the association between features within categories II and III and the risk of acidemia, but the majority of patients with these patterns will have a normal pH, which is why EFM patterns remain poorly predictive of acidemia.³⁷

Intrapartum EFM patterns have also been associated with neonatal morbidity. A large cohort study investigated the association between EFM patterns in the 30 minutes before delivery and neonatal respiratory morbidity.⁵² It found that tachycardia, marked variability,

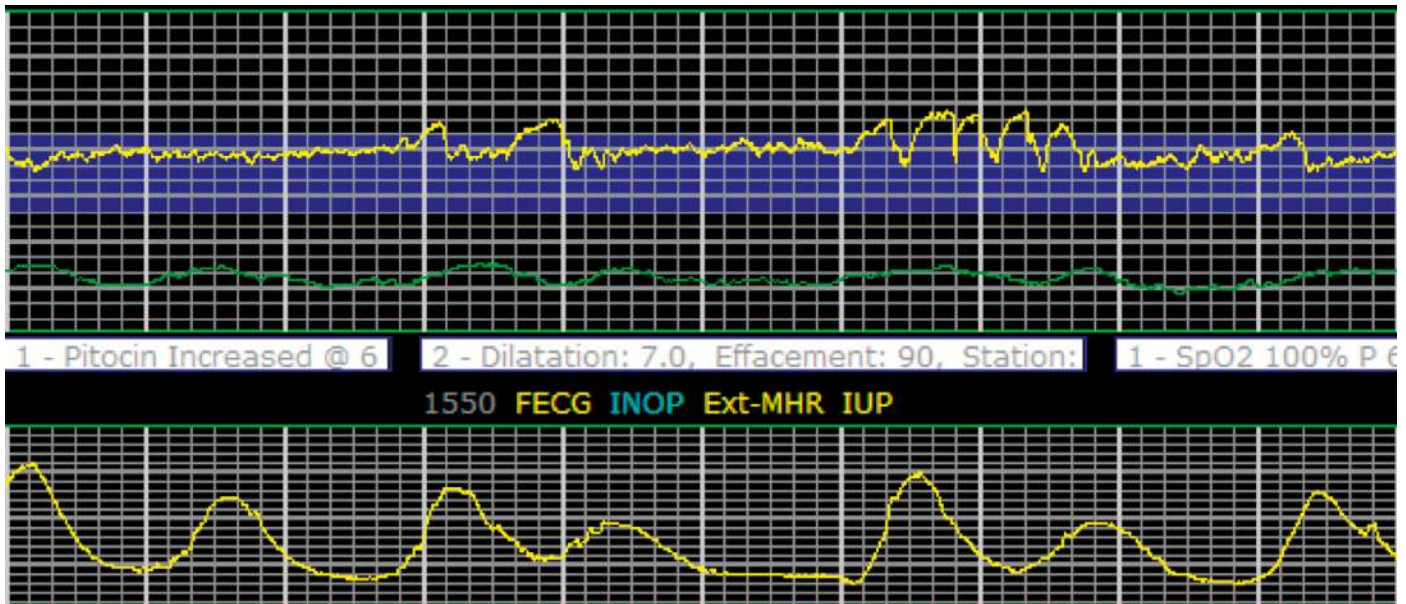


Fig. 17.29 Tachysystole Without Abnormal Electronic Fetal Monitoring.

and prolonged decelerations were associated with neonatal respiratory distress. A prospective cohort study of more than 8000 women also found an association between EFM patterns and adverse neonatal outcomes.⁵³ It was found that tachycardia and repetitive decelerations in the 2 hours before delivery were associated with neonatal morbidity at term, defined as comprising neonatal death, hypothermic therapy, respiratory distress, mechanical ventilation, meconium aspiration, seizures, or treatment for sepsis. Both studies demonstrate that features of category II and III EFM are associated with neonatal morbidities at term.

Several of these studies have also provided evidence for EFM patterns associated with normal pH. Several studies have shown that persistent category I EFM, or the presence of accelerations, is associated with a normal pH.^{46,53,54} In addition, moderate variability has also been associated with a normal pH.^{37,55} However, **it is important to recognize that neither category I nor moderate variability is 100% specific for a normal pH. It is also important to assess all of the EFM patterns together, as well as the clinical context, because it is possible to have significant risk of acidemia if other features are present and persistent despite moderate variability.**⁵³

NEWER CONSIDERATIONS

During decelerations, fetal gas exchanged is compromised. This, along with observational data in humans and animals reflecting the impact of decelerations on the risk of acidemia, has led to the concept that measures reflecting deceleration number and severity might best reflect the evolving risk for acidemia. Several authors have suggested that deceleration area, calculated by the sum of the area within decelerations (or estimates thereof) over a period of time, is more significantly correlated with acidemia.^{56,57} A retrospective cohort study examining the 30 minutes of EFM patterns before delivery found that deceleration area was more predictive of acidemia (defined as umbilical cord arterial pH ≤ 7.10) than any other NICHD-defined pattern among early-term and term infants.³⁷ In 2018, a prospective cohort study of 8580 women delivering at or beyond 37 weeks similarly found that deceleration area in the 120 minutes before delivery

was most predictive of neonatal acidemia and morbidity. They were also able to identify a threshold of deceleration area above which it would take five cesarean or operative deliveries to be performed to prevent one case of acidemia.⁵⁸ Additional work must be done to enable translation of these findings to clinical practice and determine whether the incorporation of deceleration area into the interpretation of bedside EFM improves clinical outcomes.

PERIODIC CHANGE

It is important to consider the periodic nature of EFM patterns, as they can evolve over time. Attention should be given to not only the currently observed patterns but also those that preceded them. Many transient factors—including the physiologic challenges of labor, transient hypoxia to the fetus, environmental exposures (e.g., medications), normal physiology, and maternal and fetal clinical factors—can all influence the EFM pattern. Thus it is important to assess EFM patterns in an ongoing fashion.

CLINICAL FACTORS AFFECTING ELECTRONIC FETAL MONITORING PATTERNS AND INTERPRETATION

Clinical factors have been demonstrated to affect intrapartum EFM patterns. Because these factors or exposures can independently affect the EFM patterns, it is important to consider them when intrapartum EFM patterns are being interpreted.

Magnesium Sulfate

Magnesium sulfate, used for a variety of indications in obstetrics (including seizure prophylaxis for women with severe preeclampsia, tocolysis, and neuroprotection for preterm infants), is a commonly encountered drug on labor and delivery units and is known to cross the placenta. Evidence from an animal study examining the effects of magnesium on FHR patterns suggests an associated decrease in variability and reduced accelerations.⁵⁹ An observational study of women

in labor comparing EFM patterns before magnesium initiation with the EFM patterns in the following hour found that the average baseline decreased by 2.4 beats/min but remained within the normal range. It was also found that there was an increase in the rate of minimal variability during the magnesium bolus that resolved after the bolus was complete.⁶⁰ A randomized controlled trial of magnesium compared with normal saline in nonlaboring women demonstrated a decrease in variability and average baseline after 3 hours of magnesium exposure.⁶¹ In a large retrospective cohort study, term women in the second stage of labor who were exposed to magnesium were compared with those who were unexposed. The latter had a decrease in baseline but still within the normal range, decreased variability, and fewer prolonged decelerations with no associated increase in adverse neonatal outcomes.⁶² These observational studies may be confounded by the indication for magnesium treatment, but the findings have remained consistent. **Magnesium exposure is independently associated with a small reduction in average baseline heart rate and a decrease in variability that may or may not persist after the magnesium bolus.** When these patterns are encountered in the setting of magnesium exposure, the possible contribution of the drug exposure should be considered in the differential.

Meconium-Stained Amniotic Fluid

Meconium-stained amniotic fluid is common; it is found in 12% of all laboring patients and in over 20% of patients with category II EFM patterns. In a secondary analysis of a large multicenter trial to test the efficacy of amnioinfusion (AI) in reducing the risk of meconium aspiration syndrome among women with meconium-stained amniotic fluid, it was found that abnormal EFM patterns consistent with category II or III patterns were associated with an increased risk of neonatal morbidity.⁶³ In a secondary analysis of a large prospective cohort study of laboring women at or after 37 weeks' gestation with category II patterns, Frey and colleagues found that the presence of meconium was associated with an increased risk of neonatal morbidity compared with situations where no meconium was present. They also found that the increased risk of morbidity was further increased if the meconium-stained fluid was thick rather than thin.⁶⁴ **The presence or absence of meconium-stained fluid can help to risk-stratify patients with a category II tracing for acidemia.**

Small-for-Gestational-Age Infants

These infants, defined as having a birthweight below the 10th percentile for gestational age, are a heterogeneous group made up of those who are constitutionally small and those who are pathologically growth restricted. Chronic placental insufficiency can be a cause of growth restriction and could have a potential impact on the FHR pattern. A retrospective cohort study of term infants in the second stage of labor by Epplin and colleagues⁶⁵ found significantly fewer accelerations and more late decelerations among those fetuses with growth restriction compared with those without but no differences in baseline or variability. In a secondary analysis of an intrapartum EFM trial, growth-restricted infants were less likely to have accelerations and more likely to have variable decelerations that were longer in duration (more than 60 seconds) and deeper (fewer than 60 beats/min) compared with normally grown infants in the last hour of delivery.⁶⁶ Heart rate variability, bradycardia, tachycardia, and the presence of late decelerations were similar in the two groups. The rate of neonatal composite morbidity did not differ between the groups although cesarean delivery was more commonly performed for growth-restricted infants. **Taken together, these reports imply that there are likely differences in EFM patterns among those with**

growth restriction versus those without, but the impact on infant outcomes requires further investigation.

Prematurity

Progressive development of the fetal autonomic nervous system throughout gestation influences fetal biophysical parameters, including FHR. Specifically, maturation of the parasympathetic system results in a progressive decrease in resting FHR and increase in variability with advancing gestational age.^{67,68} Accelerations are also less frequently observed during antenatal testing among preterm fetuses.⁶⁹ Unfortunately, there has been minimal research focused on the interpretation of EFM of the preterm fetus during labor. A few older studies have reported that changes in intrapartum FHR patterns such as recurrent and prolonged decelerations, decreased variability, bradycardia, and tachycardia are associated with abnormal outcomes, including acidosis and neonatal death, among preterm infants.^{70–72} This suggests that EFM interpretation in the preterm period should be similar to that at term. However, the application of these findings to modern obstetric practice is limited by differences in neonatal interventions and survival at early gestational ages, as well as variations in the definitions of fetal heart rate characteristics between the time periods of investigation and now.

ADDITIONAL TECHNOLOGIES

Because the use of intrapartum EFM has been challenged, owing to the poor predictive nature of the patterns for neonatal acidemia and morbidity, there have been efforts to explore the possibility that additional technologies might be able to add information for the clinician, ultimately improving outcomes.

Fetal Blood Sampling

Fetal blood sampling is a way to sample fetal blood directly and measure markers of acidosis. It was introduced before the use of intrapartum EFM when there were no tools for assessing intrapartum fetal well-being other than a Pinard stethoscope. The sampling of blood from the fetal scalp requires access to the fetal scalp through a dilated cervix and ruptured membranes. Because clinical indications for using fetal scalp sampling rarely arise, the tool has been used less frequently in the United States over the past two decades. In addition, as scalp sampling became uncommon, the requirements for daily validation of point-of-care testing became onerous; as a result, the tool was eventually no longer available on most labor and delivery units. Although proponents of fetal scalp sampling feel that an objective measure of fetal acid-base status by measuring either pH or lactate could reduce interventions for EFM patterns with poor specificity for fetal acidemia, there are additional concerns regarding fetal scalp sampling. First, the fetal scalp is the site of sampling because it is accessible during labor. But during fetal hypoxia, the fetus shunts blood away from noncritical organs to those more critical, such as the heart and brain. Thus sampling of the skin may not reflect the acid-base status of critical organs. Second, when there are areas of edema, such as caput medusae, sampling from those areas has demonstrated that they are more acidotic than adjacent areas with no edema from the same fetus.⁷³ Third, a prospective study of 293 patients with paired samples demonstrated that two samples from the same fetus at the same time agreed only 43% of the time,⁷⁴ reflecting the variation and questionable reliability of fetal scalp sampling. Given these considerations, existing evidence does not support the routine clinical use of fetal scalp sampling at this time.

Fetal Pulse Oximetry

Whereas intrapartum EFM provides an indirect measure of fetal acidosis and oxygenation, fetal pulse oximetry is a direct measure of fetal oxygenation. The fetal pulse oximetry probe is a noninvasive probe placed on the cheek of the fetus once the cervix is dilated and the membranes have ruptured. After reports of the accuracy of fetal pulse oximetry,^{75,76} Garite and colleagues⁷⁷ conducted a randomized controlled clinical trial of women with nonreassuring FHR tracings, testing whether the addition of fetal pulse oximetry would reduce cesarean deliveries, as well as maternal and neonatal morbidity. Similar to the findings reported from East and colleagues⁷⁸ in a randomized trial in Australia, Garite and coworkers⁷⁷ reported no difference in the rate of overall cesareans between groups but a 50% reduction in the rate of cesareans for nonreassuring fetal tracings in the pulse oximetry group. Together, these data demonstrated the safety of the addition of fetal pulse oximetry to EFM tracings to assess fetal well-being, but they conflicted with regard to the utility of fetal pulse oximetry in improving outcomes. In 2006 Bloom and colleagues⁷⁹ published the results of a multicenter randomized trial of fetal pulse oximetry conducted by the National Institute of Child Health and Human Development Maternal-Fetal Medicine Units (MFMU) Network in the United States. They found no difference in the rate of cesareans or rates of cesarean by indication, between the pulse oximetry group where the provider had knowledge of the pulse oximetry and the group that was blind to that information. The lack of significant effect of using fetal pulse oximetry to augment the interpretation of intrapartum EFM in this large multicenter trial prevented fetal pulse oximetry from entering clinical practice.

ST-Segment Analysis

Another technology that has been proposed to enhance the ability to interpret EFM patterns and identify developing fetal acidosis is ST-segment analysis. Preliminary data from animal studies suggested that fetal acidemia affects the fetal heart, leading to ST-segment elevation or depression and increased T-wave amplitude on the ECG.^{80–82} Preliminary studies in Europe suggested that the use of ST-segment analysis to augment the interpretation of EFM was associated with lower rates of acidemia, neonatal encephalopathy, and operative delivery.^{83,84} The MFMU Network in the United States then conducted a multicenter randomized trial to assess the efficacy of using ST-segment analysis to reduce neonatal morbidity. In this trial a total of 11,108 women were randomized to traditional EFM or EFM with ST-segment analysis. There were no differences in the rates of neonatal morbidity or in those of operative delivery. The results of this trial, contrary to promising preliminary reports, did not provide evidence supporting the use of ST-segment analysis to augment the clinical interpretation of EFM patterns at the bedside with a view toward reducing neonatal and operative morbidity. A meta-analysis that included the MFMU trial found a reduction in metabolic acidosis with the use of ST-segment analysis but found no reduction in operative delivery for fetal distress and no improvement in other neonatal outcomes.⁸⁵ A Cochrane meta-analysis of seven trials comparing ST-segment analysis with EFM with EFM alone—deemed to be moderate to high quality—showed no evidence that ST-segment analysis reduced the number of cesareans, severe metabolic acidosis, or neonatal encephalopathy.⁸⁶ Thus there is no evidence base for the use of this technology in current clinical practice.

Computerized Interpretation of Electronic Fetal Monitoring

Some have suggested that owing to the complexities of the patterns themselves and the variable reproducibility of visual interpretation,⁸⁷ computerized interpretation might enable the clinical use of continuous intrapartum EFM to improve neonatal outcomes. A randomized trial of 7730 women laboring with continuous EFM compared EFM alone with EFM plus computerized interpretation and real-time alerts and found no difference in the rate of umbilical cord metabolic acidosis (pH <7.05 and base excess > -12 mmol/L), which was the primary outcome.⁸⁸ Thereafter a randomized trial of more than 46,000 women across 24 hospitals in the United Kingdom and Ireland was performed.⁸⁹ Women were randomized to EFM alone compared with EFM plus computerized interpretation, including a color-coded system to designate the severity of the EFM pattern. The investigators found no difference in the primary outcome, which was a composite neonatal outcome of intrapartum stillbirth, neonatal death, encephalopathy, or admission to the neonatal intensive care unit (NICU) with evidence of morbidity. To date there are no studies demonstrating that computerized interpretation of EFM patterns reduces acidosis or meaningful neonatal morbidity compared with visual interpretation.

DIAGNOSIS AND INTERVENTIONS

Differential Diagnosis

There are several specific patterns that are components of category II or III for which possible physiologic, pathologic, and maternal exposures should be considered (Table 17.3). In the presence of tachycardia, bradycardia, a sinusoidal pattern, or minimal variability, consideration of these possible etiologies should guide clinical history taking, the physical examination, and the evaluation.

Besides removing offending agents and ruling out underlying diagnoses, a variety of interventions are recommended in response to category II and III EFM patterns by ACOG and AWHONN, among others, although the evidence to support their use is very variable in breadth and quality.

Position Change

Maternal position change is a recommended approach to categories II and III. The principle is that maternal repositioning may alter the fetal position and potentially relieve umbilical cord compression. In addition, if a woman is lying on her back, compression of the vena cava by the gravid uterus may decrease preload and subsequently reduce blood flow and oxygen delivery to the placenta and thus the fetus. One prospective observational study investigating the effect of maternal position on fetal pulse oximetry found that maternal supine positioning was associated with a lower result on fetal pulse oximetry compared with the left lateral position. Interestingly, all of the participants had normal (category I) EFM patterns. There are no data to support the use of positional change to improve category II and III patterns, but the practice is recommended by ACOG, AWHONN, and others, owing to the physiologic principles and the benign nature of the intervention.

Maternal Oxygen Supplementation

Maternal oxygen supplementation has often been used among other conservative measures to respond to category II and III patterns. This

TABLE 17.3 Electronic Fetal Monitoring (EFM) Patterns and Possible Differential Diagnoses

EFM Pattern	Physiology	Pathology	Exposures
Tachycardia	Transient sinus tachycardia	Maternal fever or infection Chorioamnionitis Fetal anemia Acidemia Maternal hyperthyroidism Fetal tachyarrhythmia	Terbutaline Stimulants (cocaine, barbiturates, amphetamines, caffeine)
Bradycardia	None	Fetal heart block Fetal structural heart disease Cardiomyopathy Maternal hypothermia Maternal hypotension Fetal hypoxia Interruption of fetal-placental gas exchange (uterine rupture, abruption, cord prolapse)	Anesthetic agents Sympatholytics Parasympathomimetics
Sinusoidal pattern	None	Fetal anemia Fetal acidosis	None reported
Minimal variability	Sleep cycle	Fetal acidemia Infection	Magnesium Anesthetic agents Narcotics Atropine

is done because some of these patterns are consistent with reduced fetal oxygenation; thus an increase in maternal inhaled oxygen was thought to increase oxygen delivery to the fetus and prevent acidemia. However, there has been increasing evidence that the use of maternal oxygen supplementation does not improve outcomes and may even be associated with harm. In one trial, Thorp and colleagues⁹⁰ found a higher rate of neonatal acidemia and lower umbilical artery pH among those randomized to oxygen compared with those given room air. Nesterenko and colleagues⁹¹ found no difference in markers of oxidative stress between those on oxygen and those on room air. But infants born to women in the oxygen group had higher rates of neonatal resuscitation in the delivery room. A larger trial of 443 women randomized to oxygen or room air found no evidence that oxygen affected umbilical artery pH.⁹² In a noninferiority randomized trial of laboring women with category II EFM patterns that required intervention, Raghuraman and colleagues⁹³ found that oxygen supplementation was not inferior to room air. They also reported that supplemental oxygen was not associated with an improvement in category II EFM features or recurrent decelerations.⁹⁴ In response to the accumulating data demonstrating no benefit of intrapartum oxygen supplementation, including the publication of a systematic review and meta-analysis that included 16 studies,⁹⁵ ACOG no longer recommends this intervention as part of the routine management of category II or III patterns in individuals with normal oxygen saturation.⁹⁶ In contrast, AWHONN continues to support its use, citing a need for additional research.⁹⁷

Intravenous Fluids

Intravenous fluid bolus administration is also part of the conservative management of category II and III EFM patterns. The underlying theory is that intravenous fluids will increase the effective circulating intravascular volume, which will then improve maternal preload and oxygen delivery to the placenta and fetus. Little data exist to describe the effect of this intervention. Simpson and colleagues⁹⁸ reported the

results of a multifactorial randomized trial, a portion of which randomized healthy women to a 500 mL versus a 1000 mL bolus and examined the effect on fetal pulse oximetry. A difference of higher fetal pulse oximetry was reported in the 1000 mL group, although this did not reach statistical significance, and the mean fetal pulse oximetry percentage in both groups was within normal limits. There are no data to guide best intravenous fluid bolus type or number of boluses. The physiologic response to regional anesthesia with the accompanying peripheral effect on the autonomic nervous system can lower peripheral vascular resistance and preload, which can affect the risk of fetal hypoxia. **Thus for maternal or fetal indications, the response to hypotension after regional anesthesia with an intravenous fluid bolus to increase the effective circulating volume is physiologically reasonable.** Prophylactic intravenous fluid administration is currently recommended either before or during administration of spinal anesthesia for cesarean delivery for the prevention of maternal hypotension, but this recommendation is not extended to those with neuroaxial anesthesia use in labor.⁹⁹ A randomized trial demonstrated that patients with a narrow pulse pressure on admission (<45 mm Hg) were less likely to experience hypotension and have a category II or III FHR pattern within 60 minutes of epidural test dose if they received a 1500 mL intravenous fluid bolus compared with a 500 mL fluid bolus at the time of epidural placement.¹⁰⁰ Additional studies focused on prophylactic intravenous fluid administration are needed to guide clinical care.

Tocolysis

Some have suggested that tocolysis may be an effective approach to category II and III patterns. The theoretical basis is that uterine hyperactivity may create an in utero environment of hypoxia and acidemia that might exceed the compensatory mechanisms of the fetus. β -adrenergic receptor agonists—extrapolating from older data when they were tested for the purpose of preterm labor tocolysis—have been used in this setting. The data to support this approach are extremely

limited. In a systematic review and meta-analysis of eight small trials, the findings suggested that tocolysis with β -adrenergic receptor agonists may reduce tachysystole and improve FHR tracings compared with no treatment, but there were no differences in outcomes such as Apgar scores or umbilical artery pH.¹⁰¹ Overall, the authors concluded that there was insufficient evidence to determine the effect of tocolytic therapy because the quantity of available data was scant and the quality of the studies was deemed low/moderate. Nevertheless, this clinical practice of tocolytic administration in the setting of concerning FHR patterns is not uncommon. **Terbutaline should be used with caution, particularly in women with comorbidities, given that the data to support its use in this setting are poor and that there are known maternal and fetal side effects.**¹⁰²

Amnioinfusion

There is a better evidence base for the initiation of an AI in the setting of recurrent variable decelerations. An AI is accomplished by infusing normal saline into the uterus via intravenous tubing connected to an intrauterine pressure catheter. Following early studies in animals demonstrating that fluid replacement in the setting of oligohydramnios could decrease variable decelerations, a small observational study in humans revealed the same.¹⁰³ A randomized trial of 96 women in the first stage of labor with recurrent variable decelerations demonstrated that 51% of those who received an AI had deceleration resolution compared with 4.2% of those who did not receive such an infusion.¹⁰⁴ Other studies have also demonstrated benefit from AI in terms of both reduction of recurrent decelerations and a decreased likelihood of cesarean delivery for fetal indications.¹⁰⁵ Most trials tested AIs with a 250 mL to 500 mL bolus followed by a continuous infusion.¹⁰⁶ Only one small trial has tested the approach of intermittent infusion,¹⁰⁶ but there have been no other studies to confirm its findings or to support that approach. Another small trial of 53 women randomized patients to an AI with lactated Ringer solution versus normal saline and found no difference in neonatal electrolyte profiles.¹⁰⁷ But the majority of trials testing the efficacy of AI used normal saline. None of the published trials have demonstrated risks of an AI, but there are known risks extrapolated from the general IUPC literature to consider, including a slightly increased risk of maternal febrile morbidity.¹⁰⁸ **A Cochrane meta-analysis summarized the available data, highlighting that in the setting of recurrent variable decelerations, initiation of an AI is a reasonable evidence-based strategy to reduce recurrent decelerations and the likelihood for cesarean delivery for category II or III EFM patterns.**¹⁰⁵

Fetal Scalp Stimulation

FHR acceleration is an EFM pattern that is strongly associated with a normal fetal pH. In the absence of spontaneous accelerations, fetal scalp stimulation can be used to elicit an acceleration and convey a similar level of confidence regarding the absence of fetal acidemia. The concept of fetal scalp stimulation first emerged when it was observed that increases in FHR, often accelerations, occurred at the time of fetal scalp sampling, which behaved as a stimulus. Therefore Clark and colleagues¹⁰⁹ demonstrated that when fetal scalp stimulation with an Allis clamp was followed by an acceleration, it was 100% sensitive for a scalp pH of equal to or greater than 7.19. Seeking a less abrasive method to elicit reassurance of a normal pH, Elimian and colleagues¹¹⁰ studied 103 term infants and found that digital stimulation followed by an acceleration was 100% sensitive for a scalp pH equal to or greater than 7.20. Subsequent larger studies have demonstrated that although less than 100% sensitive, accelerations following scalp stimulation are significantly associated with a normal pH.¹¹⁰ **Scalp**

stimulation represents a low-risk, inexpensive, and readily available tool to assess for a normal fetal pH in the setting of category II and III patterns.

Clinical Algorithms

The central limitation of the three-tiered category system is related to the grouping of a diverse collection of EFM patterns within category II. While some category II patterns may be associated with acidemia and neonatal morbidity, the majority are not. Experts have proposed algorithms to approach category II EFM patterns. To date, none have been tested in a randomized trial or have been shown to reduce neonatal morbidity, but they do represent expert opinion. One algorithm was suggested by Clark and colleagues¹¹¹ to manage category II FHR patterns. Their expert opinion was that perhaps a contributor to the conflicting evidence regarding the effectiveness of intrapartum EFM was the lack of standardized approaches to management and intervention. With that in mind, they proposed a standardized approach to category II patterns with consideration of the presence or absence of moderate variability with or without accelerations and/or recurrent decelerations. On the basis of observations of these patterns and the phase of labor, they recommend a course of action that is either conservative management or delivery. Although certainly practical—and the premise that routine approaches to category II EFM patterns would reduce variation in interventions is reasonable—this algorithm and others proposed to date have not been tested. That is, there have been no trials to investigate the management of category II patterns based on this or other protocols compared with usual practice and their impact on clinically meaningful outcomes. Thus there remains an urgent need for the development of effective and validated methods to risk-stratify patients with category II EFM patterns. In the meantime, obstetric providers will continue to rely on the identification of individual or combinations of features of EFM within the category II group associated with adverse neonatal outcomes in observational studies (e.g., minimal variability with recurrent late decelerations) without a more uniform approach to the interpretation and management of this specific EFM category.

TRIAL OF LABOR AFTER CESAREAN CONSIDERATIONS

Continuous intrapartum EFM is particularly important during TOLAC because of the risk of uterine rupture. Although uterine rupture is a rare event, abnormalities in the FHR pattern often precede or accompany uterine rupture. Certainly uterine rupture should be considered in the differential for any patient attempting vaginal birth after cesarean who develops a category II or III EFM pattern. Because rupture is a rare event, most commonly the patterns will arise for the same reasons as in any other laboring patient. But rupture should always be considered and ruled out. Specifically, there are reports that repetitive variable or prolonged decelerations and/or fetal bradycardia are the most commonly seen patterns preceding or during a uterine rupture.^{112,113}

CONSIDERATION FOR FETAL ABNORMALITIES

Existing data describing the use of intrapartum EFM and the interpretation of patterns are largely based on observational data among nonanomalous term infants. The positive and negative predictive value of EFM patterns for acidemia and/or morbidity among infants with anomalies and preterm infants remains poorly studied. In addition, fetal arrhythmias, either tachyarrhythmias or bradyarrhythmias,

typically necessitate cesarean delivery because continuous monitoring cannot be accomplished and/or the ability to interpret fetal well-being is compromised.

There are clinical circumstances in which decisions are made not to monitor the fetus during labor. Most commonly this is in the setting of life-limiting fetal anomalies or conditions where avoiding the maternal risks associated with urgent cesarean in response to abnormal FHR patterns is favored over the potential avoidance of an intrapartum fetal demise. These decisions are complex, often involving multidisciplinary counseling, and are patient and family centered. Additional details are beyond the scope of this chapter.

UMBILICAL CORD GAS ANALYSIS

Assessment of the fetal acid-base status at the time of delivery has become increasingly important. Umbilical cord pH decreases when persistent hypoxia causes a shift to anaerobic metabolism and hydrogen ions accumulate and overwhelm the capacity of the fetal buffer system.¹¹⁴ Because the sensitivity and specificity of EFM patterns has remained relatively poor, the best objective measure and the gold standard of fetal acid-base status is to directly measure it from one of the umbilical arteries and compare the result to normative values (Table 17.4). Many studies have demonstrated that umbilical cord arterial pH is strongly associated with neonatal morbidity and mortality. A meta-analysis of 51 studies demonstrated that an arterial pH below 7.00 was strongly associated with neonatal morbidity (OR, 12.5; 95% CI, 6.1 to 25.6) and mortality (OR, 6.1; 95% CI, 0.9 to 41.6) and that an arterial cord pH below 7.10 was also significantly associated with morbidity (OR, 2.4; 95% CI, 1.3 to 4.2) and mortality (OR, 7.1; 95% CI, 3.3 to 15.3).¹¹⁵ The components of the umbilical cord gas can also be evaluated to try to further qualify acidemia as respiratory, metabolic, or mixed. Respiratory acidosis is thought to reflect a more acute process with an elevation in pCO₂, whereas a metabolic acidosis reflects chronic impaired oxygenation with evidence of anaerobic metabolism reflected by an elevated base excess (Table 17.5).

The ACOG recommends obtaining paired (both arterial and venous) umbilical cord gases if there is any anticipation of neonatal depression or anticipation of morbidity.¹¹⁶ The rationale for obtaining paired umbilical cord gases is to ensure that the reported result for the arterial gas is, in fact, arterial. Some experts and centers have moved to universal paired umbilical cord gas sampling.^{27,117} The rationale behind that approach is twofold. First, acidemia is rare, even in the setting of a category II EFM tracing, which is what is seen in labor most of the time; thus most cord gases are normal even when preceded by those patterns. The second rationale is to identify acidemia in an unexpected case because even a category I tracing is not 100% specific for a normal pH, thus enabling the identification of infants who might benefit from additional treatment, such as serial neurologic assessments for therapeutic hypothermia. When an

TABLE 17.4 Normal Umbilical Cord Blood Values

Vessel	pH	PCO ₂ (mm Hg)	PO ₂ (mm Hg)	Base Deficit (mmol/L)
Artery	7.2–7.3	45–55	15–25	<12
Vein	7.3–7.4	35–45	25–35	<12

TABLE 17.5 Categories of Umbilical Artery Acidemia

Value	Respiratory	Metabolic	Mixed
pH	<7.20	<7.20	<7.20
PCO ₂	Elevated	Normal	Elevated
Base deficit	<12 mmol/L	≥12 mmol/L	≥12 mmol/L

umbilical cord gas sample, which is taken from a clamped segment of umbilical cord within 30 minutes of delivery, is being obtained, it can be more challenging to obtain an arterial sample than a venous sample, owing to the caliber of the vessel and the amount of blood in it. This challenge means that there are clinical cases where only a venous gas can be obtained.

When paired samples are not obtained, it can also lead to a single umbilical cord gas result and the inability to determine whether it is venous or arterial. One study at a center where universal paired umbilical cord sampling is performed examined threshold values of the venous umbilical cord blood gas that would rule out acidemia among more than 11,000 births.¹¹⁸ It was found that a venous pH of 7.23 or greater meant that the probability of an arterial pH below 7 would be 1% or less and also that a venous pH of 7.29 or greater would mean a probability of 1% or less of an arterial pH below 7.10. This was particularly important because in this cohort, despite the universal cord gas policy, only 62% had paired validated samples. Thus the ability to predict the possibility of acidemia from a venous sample is important.

Umbilical cord arterial lactate can also be measured. Because lactate is a direct end product of anaerobic metabolism, some studies have suggested that it is more sensitive and specific than pH for identifying neonates at risk for morbidity.¹¹⁹ Umbilical cord lactate is also thought to be primarily fetal in origin, as opposed to placental or maternal.¹²⁰ A large prospective cohort study of an unselected term infant population compared umbilical artery pH with lactate for predicting neonatal morbidity. This study found that lactate was superior to pH and specifically that umbilical cord lactate greater than 3.90 mmol/L had a sensitivity of 83.9% and a specificity of 74.1% for neonatal morbidity.¹²¹ Similar to the ability to use a single venous pH sample to predict arterial pH, a large prospective study demonstrated that a cutoff of venous lactate of 3.4 mmol/L was 91.3% specific for arterial lactic acidemia and equivalent to arterial lactate for predicting neonatal morbidity.¹²² However, many centers do not have the ability to measure lactate, in which case traditional umbilical cord gas analysis is advised. **Traditional parameters, such as pH and calculated base excess, are still currently used to screen infants for morbidity, including encephalopathy.**¹¹⁶

CONSIDERATIONS FOR ELECTRONIC FETAL MONITORING: REVIEW

Because scientific work to improve the clinical use of intrapartum EFM is emerging and ongoing, a few evidence-based principles are important to consider. First, studies examining the association and predictive ability of EFM patterns for acidemia can be subject to potential selection bias if umbilical cord gases are obtained on the basis of obstetric or neonatal criteria instead of being universally obtained. In addition, extraction of the EFM interpretation from the medical record can introduce bias, as this can frequently be interpreted and charted

after clinical events have occurred, the outcome is known, or even after subsequent patterns have been observed. Similarly, review and interpretation of the EFM patterns themselves for research purposes that are not blind to clinical factors and outcomes can be equally biased. Finally, some investigators are using computerized extraction and/or interpretation, but these techniques have yet to be validated or shown to improve outcomes; they also call into question the generalizability of the findings. Important work is being conducted to improve the use of EFM at the bedside, but emerging evidence should be evaluated with these considerations in mind.

MINIMIZING ERRORS

It is important to distinguish the FHR from the maternal heart rate and to make every effort to ensure that the heart rate pattern being observed and interpreted is, in fact, fetal. As reviewed earlier, because external FHR monitors use ultrasound technology, it is possible that the signal detected is actually pulsation from a maternal vessel as opposed to movement of the fetal heart. Although distinguishing between the two may be straightforward at most points during labor, since the normal FHR is between 110 and 160 beats/min and the normal maternal heart rate is between 80 and 100 beats/min, there are circumstances when the distinction may be more challenging (Fig. 17.30). During fetal decelerations and certainly fetal bradycardia, the FHR can easily be confused with the maternal rate. Additionally, during periods of maternal tachycardia occurring due to exertion, pain, or fever, the maternal heart rate can rise into the normal fetal range and can be difficult to distinguish from the FHR. A helpful practice is to use maternal pulse oximetry, which plots maternal heart rate on the same paper tracing or digital display as the FHR. In addition, when clinically appropriate, use of a fetal scalp electrode to acquire the fetal signal directly can further help to distinguish between the two heart rates. This consideration is especially important during the second stage of labor, when the combination of maternal tachycardia due to exertion and fetal decelerations with contractions is commonly encountered.

SUMMARY

The use of continuous intrapartum EFM is widespread. Although early trials of labor management comparing intrapartum EFM to intermittent auscultation did not provide conclusive support for its use, subsequent epidemiologic studies have suggested that the use of EFM has been associated with a decrease in intrapartum fetal death and neonatal morbidity and mortality. In addition, several studies have demonstrated a significant association between EFM patterns with acidemia and morbidity, as well as other patterns associated with normal pH. However, a consistent finding of an increase in operative delivery has been demonstrated across studies, highlighting the need for careful clinical interpretation and for additional guidance regarding the meaning of complex patterns.

When EFM patterns are being interpreted, visual evaluation should be used at the bedside, with appreciation for periodic changes in patterns and the clinical context. The accepted NICHD nomenclature should be used for clinical communication and documentation, as well as in future research. Category I has been consistently and significantly associated with normal pH. Although all of the interventional strategies are recommended alone or in combination in response to category II and III EFM patterns, the speed with which patterns are recognized and responded to and the amount of time allowed until resolution differ. Category II encompasses a wide range of patterns that vary in their strength of association with acidemia and are frequently encountered in the course of most labors. In comparison, category III represents a small group of rarely encountered patterns that, as a group, are strongly correlated with acidemia. Thus although initial measures may be the same or similar, category III patterns must be resolved rapidly or prompt delivery will be necessary.

This does not mean that delivery necessarily must be accomplished by cesarean section. If a patient is in the second stage of labor, often spontaneous vaginal delivery or operative vaginal delivery is appropriate and can result in a more rapid delivery than cesarean delivery.

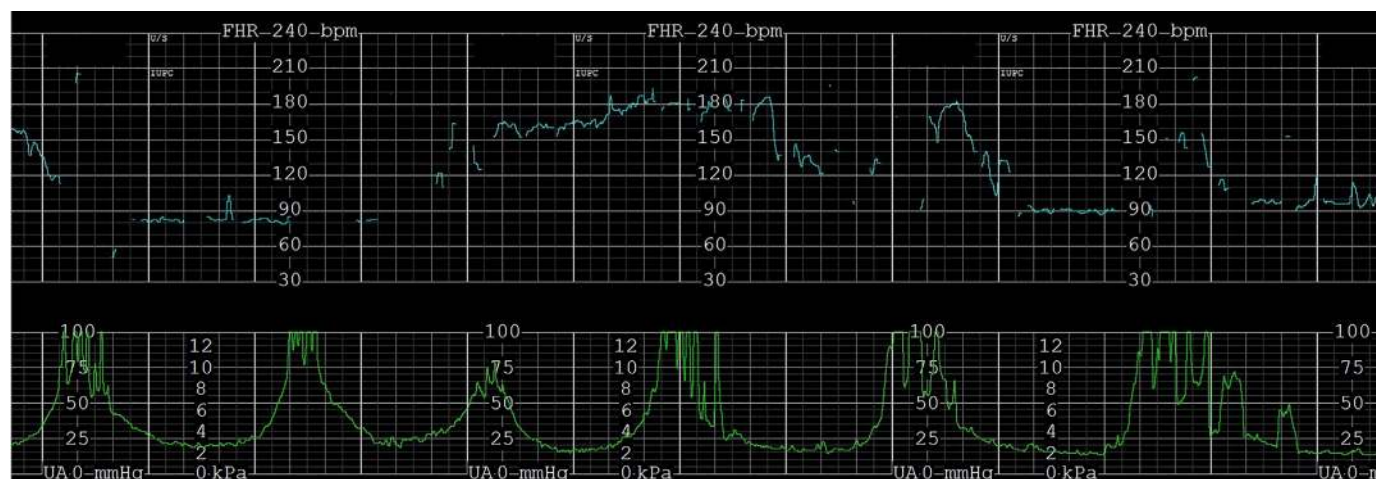


Fig. 17.30 Signal Ambiguity.

KEY POINTS

- Intrapartum fetal monitoring to assess fetal well-being during labor and delivery is a key component of intrapartum management; more than 80% of laboring patients in the United States undergo intrapartum EFM.
- Uterine monitoring is an important part of intrapartum fetal monitoring because it enables the interpretation of EFM patterns in the context of uterine activity.
- Intrapartum EFM patterns have been associated with acidemia and short-term neonatal morbidity.
- The presence or absence of meconium-stained fluid can help to risk-stratify patients with a category II tracing for acidemia.
- Neither ST-segment analysis nor fetal pulse oximetry has been shown to assist in the interpretation of EFM patterns for improved outcomes.
- In the setting of recurrent variable decelerations, initiation of an AI is a reasonable evidence-based strategy to reduce recurrent decelerations and the chance for cesarean section for category II or III EFM patterns.
- When EFM patterns are being interpreted, visual evaluation should be used at the bedside with appreciation for periodic changes in patterns and the clinical context. The accepted NICHD nomenclature should be used for clinical communication and documentation, as well as in future research.
- Scalp stimulation represents a low-risk, inexpensive, and readily available tool to assess for a normal fetal pH in the setting of category II and III patterns.
- The ACOG recommends obtaining paired (both arterial and venous) umbilical cord gases if there is any anticipation of neonatal depression or morbidity.
- It is important to distinguish the FHR from the maternal heart rate and to make every effort to ensure that the heart rate pattern being observed and interpreted is, in fact, fetal. A helpful practice is to use maternal pulse oximetry, which plots maternal heart rate on the same paper tracing or digital display as the FHR.

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